Queryable Gaussian View Index: Text Retrieval and Selective Camera Localization

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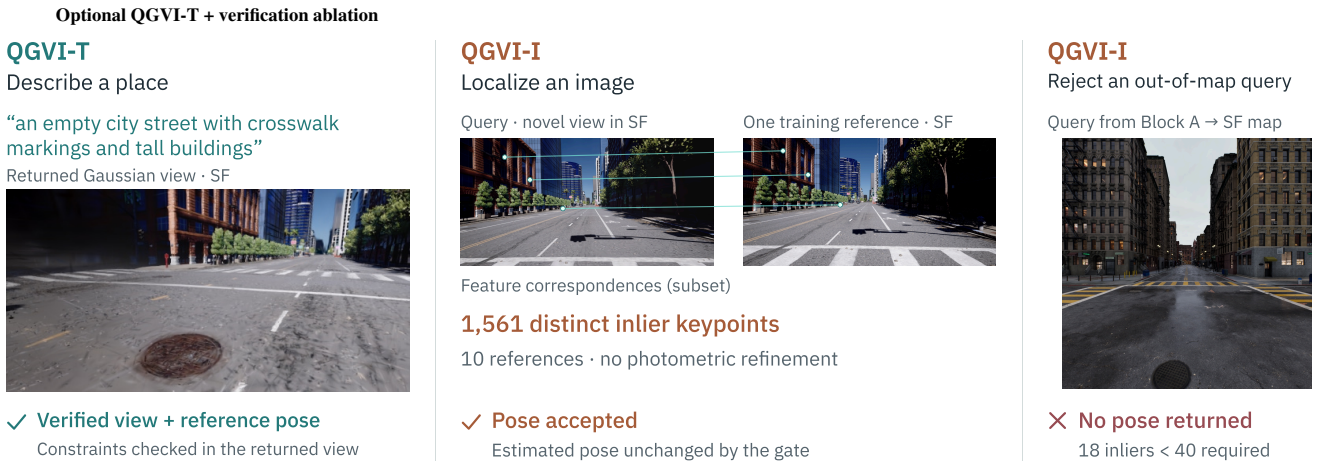


Figure 1. **Queryable Gaussian View Index (QGVI).** A retrieved view and its stored reference pose are shown with the optional text-verification checks (left). QGVI-I accepts a novel Gaussian-rendered query (middle) and rejects a query from another scene (right), preserving accepted poses. These saved examples use synthetic SF and Block A data. Middle: one of ten pooled references and a subset of feature correspondences are shown.

Abstract

We present Queryable Gaussian View Index (QGVI) for querying existing Gaussian scene assets and their associated calibrated reference data. Text-view retrieval and selective RGB camera localization return poses in the asset's coordinate frame. QGVI-T turns Gaussian-rendered views into a language-searchable index of scene locations, binding structured captions to known camera poses and place identifiers. Semantic and lexical retrieval select views from distinct places without online rendering, VLM calls, or a scene-specific pose network. The image route, QGVI-I, filters LoGS-based geometric estimates using distinct inlier query keypoints. It omits photometric refinement and preserves accepted pose coordinates. On nine development scenes, QGVI-T achieves 58.3% Top-1 and 78.3% Top-5 joint relevance-and-validity success, compared with 26.1% and 34.6% for our CityLoC reimplementation with refinement. QGVI-I achieves 97.3% scene-mean pose success at 1 m/2°, with pooled false-acceptance and false-rejection rates of 0.707% and 1.142%. A paired cached evaluation retains all 1,775 initializer poses correct at this threshold. Recorded median text query latency is 0.020 s. Matched controls show competitive BM25 retrieval and slightly lower false acceptance for thresholded full LoGS; grid augmentation suggests modest gains on novel-view-derived descriptions. We report model-judged semantic success separately from source-camera recovery, with source-verified or estimated scales for geometric evaluation.

1. Introduction

We study how to locate viewpoints within an existing Gaussian scene asset from a text description or an RGB observation. The reconstructed asset and its associated calibrated reference data are supplied inputs. Both query modalities seek camera poses in the asset's coordinate frame, with different conditions on a valid answer. A description may admit several viewpoints; an RGB observation constrains the camera that captured it. A retrieved view may omit a required attribute, and a localizer may return a pose for an image from another scene (Fig. 1).

For semantic scene querying, a description specifies conditions on a view without necessarily identifying its source camera. Several distant poses may satisfy the same conditions. A continuous pose distribution can represent this ambiguity, but estimating that distribution is not required to return a bounded set of satisfying views. We therefore evaluate semantic satisfaction separately from source-camera recovery. This formulation retains the need for geometric pose accuracy when the query is an RGB image.

Gaussian splatting supports fast rendering of reference images at known poses [18]. CityLoC uses Gaussian scenes for text-conditioned pose localization [26]; LoGS combines geometric initialization and render-based refinement for visual localization [8]. Queryable Gaussian View Index (QGVI) provides semantic view access and geometric camera localization within the supplied asset. Its routes use separate reference assets and solvers: QGVI-T retrieves Gaussian-rendered views with stored poses, while QGVI-I estimates camera poses from original RGB references and sparse 3D landmarks. Their common scene-grounding formulation supports a shared accounting of correct returns, incorrect returns, and coverage, with modality-specific correctness criteria.

Our primary contribution, QGVI-T, makes the supplied Gaussian asset searchable through a posed view index. Virtual cameras sample observations beyond the captured training views. Each index entry links a structured description to visible evidence, a camera pose, and an identifier shared by views at the same sampled camera center. Sentence-level semantic retrieval and lexical retrieval jointly address paraphrases and explicit terms, including visible text. Place-aware selection prevents several views from one indexed location from exhausting the five-result budget. Rendering and captioning are performed offline; a query searches the index without loading the Gaussian model, calling a VLM, or training a scene-specific pose network. An original-image-only control isolates view-bank augmentation: confirmed retrieval gains are small and concentrated in descriptions authored from novel views, with uncertainty limiting a general improvement claim.

The complementary image route, QGVI-I, applies an acceptance gate to the LoGS geometric initializer, omitting photometric refinement. The gate counts distinct inlier query keypoints, avoiding repeated support from one feature matched across several references. A threshold selected on Block A and SF validation queries is frozen across nine test scenes. Every accepted pose retains its original coordinates. We evaluate separately whether the gate rejects any estimate that satisfies the pose-accuracy thresholds.

Our contributions are:

- A posed view index (QGVI-T) that connects structured language to Gaussian-rendered scene observations, combining semantic and lexical retrieval with selection across distinct places.
- Selective camera localization (QGVI-I), which filters LoGS-based geometric estimates using distinct inlier query keypoints and preserves accepted pose coordinates.
- A nine-scene evaluation with a shared accounting of correct returns, coverage, and accepted-result precision, while distinguishing semantic satisfaction, source-camera recovery, and RGB pose accuracy.

On this benchmark, QGVI-T attains 78.3% Top-5 success with a recorded 20 ms median query latency. An optional verification ablation improves Top-1 success but reduces Top-5 success. The RGB gate retains every previously correct pose in the evaluated cohort. These findings concern model-judged development data; they do not guarantee correct-pose retention on unseen queries.

2. Related Work

Text-conditioned localization. Text-conditioned localization connects a description to a position or pose in a scene. Text2Pos and Text2Loc localize within point-cloud maps through coarse-to-fine inference [20, 44]; PMSH models uncertainty when descriptions only partially match submaps [12]. CityLoC [26] instead learns a text-conditioned distribution over camera poses using diffusion and rendered candidates for refinement. Such distributions accommodate multiple plausible poses. Our semantic query task instead asks for a bounded set of satisfying views; source-pose error alone cannot credit every valid alternative. This task distinction does not establish that discrete retrieval universally outperforms continuous estimation.

Semantic scene representations also support spatial queries. CLIP [30] aligns image and text representations, and OpenScene aligns point features with image and text embeddings [28]. LERF, LangSplat, and Feature 3DGS extend language or semantic features to renderable scene representations [19, 29, 47]. QGVI-T searches posed Gaussian views using BM25 [32] and MiniLM-based sentence embeddings [31, 43], preserving explicit terms alongside semantic similarity and selecting distinct indexed places. Its semantic evaluation measures description satisfaction at returned views; source-camera recovery is evaluated separately.

Geometric localization. Structure-based methods estimate pose from image-to-map correspondences, with retrieval and local matching determining the available geometric evidence. Active Search prioritizes direct 2D–3D matching [36]; hierarchical localization retrieves candidate places before local matching [33]. InLoc further combines dense matching with synthesized-view verification [40].

Global descriptors, including NetVLAD, MixVPR, and SALAD, retrieve reference images [1, 3, 16]. Local representations range from SIFT and its RootSIFT normalization [2, 25] to learned detectors and descriptors such as SuperPoint and D2-Net [10, 11]. SuperGlue and LightGlue assign sparse feature matches [24, 34]; LoFTR establishes detector-free correspondences [39]. For pose recovery, COLMAP supplies sparse landmarks [37], while PnP solvers such as EPnP [21] and robust consensus estimation [13] handle noisy matches. Scene-coordinate regression learns image-to-scene associations directly: DSAC makes robust estimation differentiable [6], and ACE accelerates scene-specific training [7]. QGVI-I adopts NetVLAD, SuperPoint, SuperGlue, and SfM landmarks through the LoGS initializer [8]. Its added acceptance rule counts distinct inlier query keypoints, avoiding repeated support across references, and leaves the accepted pose fixed.

Rendering and pose refinement. Pose refinement can use rendered appearance or feature alignment. NeRF represents continuous radiance fields [27]; iNeRF refines an initial pose using rendered-image residuals [45]. Feature-based alternatives include PixLoc's learned multi-scale alignment [35] and MCLoc's pretrained features with particle filtering [42]. 3D Gaussian Splatting supports differentiable novel-view rendering [18]. GSLoc differentiates image-alignment error with coarse-to-fine blurring [5]; LoGS adds affine brightness correction and informative-pixel masks to photometric refinement [8]. GSplatLoc grounds keypoint descriptors in Gaussian maps [38], and SplatHLoc combines rendered and image-derived features at different matching stages [41]. 6DGS estimates pose from image–ray bindings without requiring an initial pose or iterative pose optimization [4]. Rendering supplies QGVI-T's candidate views. QGVI-I uses original RGB references and SfM landmarks, applying its acceptance rule after geometric initialization without photometric refinement.

Selective prediction. Selective classification separates prediction from withholding an output, exposing a risk–coverage trade-off [15]. For localization, accepting an out-of-map image, rejecting an in-map image, and accepting an inaccurate in-map pose are distinct outcomes. QGVI-I selects its threshold subject to retaining validation-correct poses and holds the underlying estimates fixed in the paired gate ablation. The optional QGVI-T verification ablation withholds views whose required visual constraints cannot be verified. These decisions are evaluated separately from view relevance and pose accuracy.

3. Method

3.1. Task: Querying a Gaussian Scene Asset

The input scene M comprises an existing Gaussian reconstruction and its associated calibrated reference data. The task is to return camera poses within this asset's coordinate frame that satisfy a query. For query q of modality $m \in \{\mathrm{T}, \mathrm{I}\}$, define the valid-answer set

$$\mathcal{A}_m(q, M) = \{T \in SE(3) : C_m(q, T; M) = 1\}. \quad (1)$$

Here T is a scene-frame camera pose. For text, C_{T} requires a usable view satisfying the description; several distant poses may qualify. For RGB, C_{I} requires an in-map query and agreement with its source pose within the evaluation thresholds. Out-of-map RGB queries have an empty valid-answer set. This set defines the evaluation target; neither route computes it exhaustively. QGVI-T retrieves up to five indexed candidate views with stored poses. QGVI-I estimates one continuous camera pose or rejects the query. Default QGVI-T has no calibrated abstention rule.

The routes use different reference assets (Fig. 2). QGVI-T stores rendered images from frozen Horizon-GS reconstructions [17, 18], with captions, camera-to-world poses, and place identifiers. QGVI-I uses original training RGB images and sparse landmarks triangulated from feature matches with fixed training cameras [37]; its inference requires no Gaussian model. We use these assets with the LoGS initializer [8]. Supplementary Sec. B gives implementation settings.

3.2. QGVI-T: Language-Indexed View Retrieval

Posed index construction. We render a deterministic lattice of camera centers near the training cameras, with sampled orientations and training-camera intrinsics. Within each scene and camera domain, views sharing a center share a place identifier. The index therefore includes virtual observations beyond the captured views, each with an explicit camera pose.

Caption retrieval. DeepSeek describes each rendered reference offline using structured fields for objects, attributes, counts, relations, and visible text. The prompt prohibits inferring hidden objects or camera metadata; the uncertainty field is excluded from the index. Reference images and poses remain fixed.

Let c_{ij} be caption sentence j for view i and $e(\cdot)$ the normalized all-MiniLM-L6-v2 embedding [31, 43]. For query q, the semantic score is $d_i(q) = \max_j e(q)^\top e(c_{ij})$. We

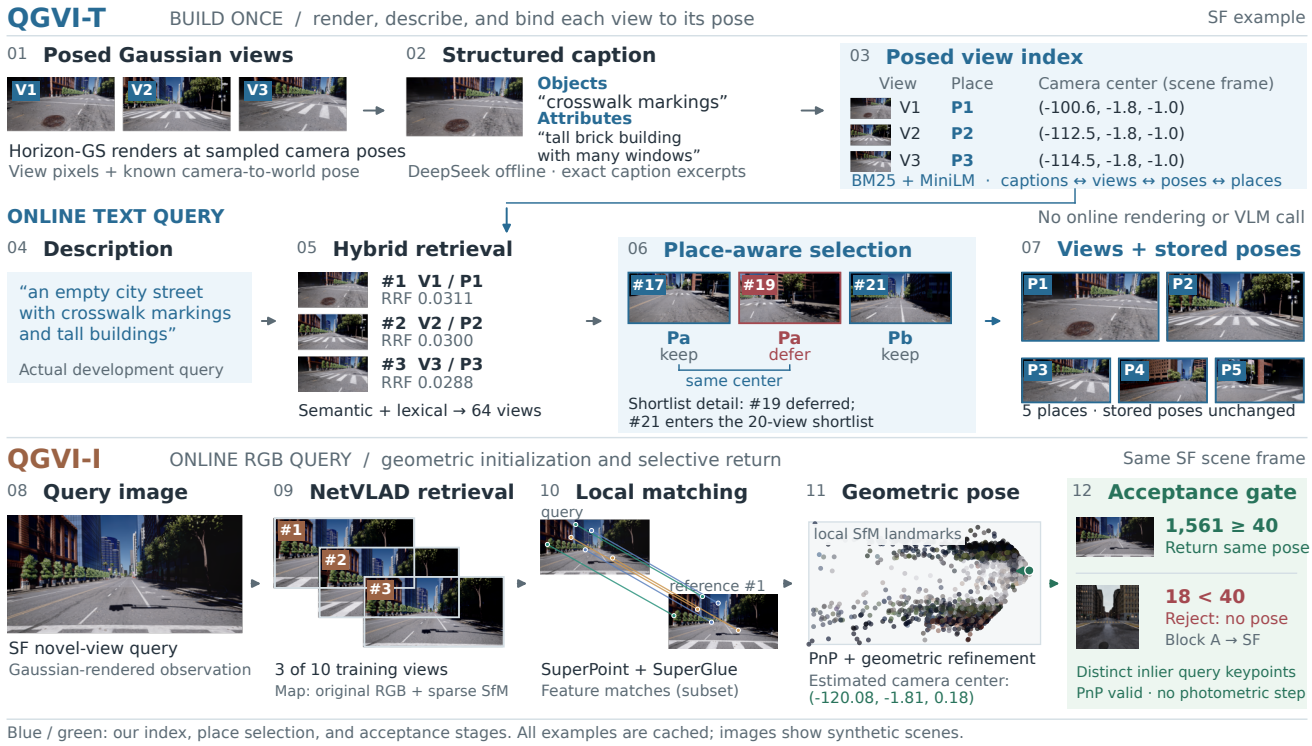


Figure 2. **QGVI illustrated with cached examples.** Top: Gaussian renders, exact caption excerpts, and posed index records. Middle: a real text query, hybrid ranks, place-aware selection, and five returned views. Raw ranks 17 and 19 share a camera center; deferring rank 19 admits rank 21 to the 20-view shortlist. View/place labels are display aliases. Bottom: an SF query, actual NetVLAD references, a subset of cached feature matches, local SfM landmarks, and acceptance decisions. The green marker denotes the estimated camera; 1,561 distinct inlier query keypoints retain its pose, while 18 reject a separate Block A distractor. Blue and green highlight our index, selection, and gate. All images depict synthetic scenes; optional text verification is outside this default pipeline.

fuse semantic and whole-caption BM25 [32] rankings by reciprocal-rank fusion [9]:

$$s_i(q) = \frac{1}{60 + r_i^{\text{sem}}} + \frac{\mathbf{1}[b_i(q) > 0]}{60 + r_i^{\text{lex}}}, \tag{2}$$

where $b_i(q)$ is the BM25 score and ranks start at one. A zero BM25 score contributes no lexical term. We retain 64 candidates and prioritize the first occurrence of each place identifier. From the first 20 views in this ordering, we return at most five distinct places with their unchanged stored poses. QGVI-T uses no online visual verifier. Its result count reflects available indexed places, not a certificate of semantic answerability.

Optional verification ablation. QGVI-T + verification checks a 20-view shortlist for satisfaction of all required textual constraints, then returns up to five distinct places or no result. Verification is excluded from the default route; the complete rule appears in Supplementary Sec. G.

3.3. QGVI-I: Pose-Preserving Selective Localization

Geometric initialization. For image I with intrinsics K, the LoGS initializer retrieves ten training images with NetVLAD [3] and matches SuperPoint features [10] using SuperGlue [34]. Reference feature indices connect matches to triangulated landmarks. Correspondences from all references form one PnP problem without covisibility clustering. PnP-RANSAC [13] uses a 12-pixel reprojection threshold and is followed by geometric pose refinement. QGVI-I omits LoGS's photometric refinement.

Distinct geometric support. Matches from several references can repeat the same query feature. At the resulting camera-to-world pose T_0, we therefore count each query keypoint once, even when it matches several references or competing landmarks. For keypoint u_j, let $\mathcal{P}_j^+(T_0)$ contain its matched landmarks with positive depth. Selecting the

smallest reprojection residual gives

$$n_{\mathrm{u}}(T_0) = \sum_j \mathbf{1}\left[\min_{X \in \mathcal{P}_j^+(T_0)} \|\pi_K(T_0^{-1}X) - u_j\|_2 \le 12\right]. \tag{3}$$

The minimum of an empty set is infinity, and π_K is calibrated projection. We recompute this count at the saved pose without rerunning the solver. It measures distinct query-keypoint support rather than the number of correspondence rows.

Acceptance calibration. We select one policy on 490 Block A and SF validation queries, equally split between in-map and out-of-map examples. A predefined grid considers minimum distinct inliers, inlier fraction, and image coverage. Feasible policies retain every validation pose correct at either 1 m/2° or 5 m/5°; selection prioritizes out-of-map rejection and then in-map retention. The selected rule requires successful PnP and at least 40 distinct inliers, with fraction and coverage thresholds zero. Frozen for all nine test scenes, it returns

$$\begin{aligned} a(I) &= \mathbf{1}[\text{PnP succeeds}]\mathbf{1}[n_{\mathrm{u}}(T_0) \ge 40], \\ \widehat{T}(I) &= \begin{cases} T_0, & a(I) = 1, \\ \varnothing, & a(I) = 0. \end{cases} \end{aligned} \tag{4}$$

Retrieval-camera fallbacks are rejected. For accepted queries, the gate returns T_0 without arithmetic or refinement, so its coordinates are unchanged. Whether the rule retains correct poses beyond validation is measured in Sec. 4, which also compares full LoGS with photometric refinement.

4. Experiments

Across nine scenes, we evaluate text-view retrieval, RGB pose accuracy, rejection errors, and query latency. Supplementary Secs. A and C provide historical comparisons and protocol details.

4.1. Experimental Setup

Datasets and reference views. The nine-scene benchmark combines Block Small and Block A from MatrixCity [22], NYC and SF from UC-GS [46], and Citysample, Colosseum, Elvenruin, Park, and Road from the Horizon-GS dataset [17]. Seven scenes are synthetic; Park and Road contain real captures. We use frozen Horizon-GS reconstructions across all nine scenes to generate reference views and novel-view queries. The text bank contains 30,522 rendered views. The source split has no exact file overlap, but some evaluation observations contributed to reconstruction.

Query cohorts. The main text cohort contains 180 DeepSeek-authored descriptions, 20 per scene: 74 from original test images and 106 from novel Gaussian renders; six are unsupported by their source images. Earlier 176-query and 135-query cohorts remain separate.

RGB testing uses 1,839 in-map and 1,839 out-of-map queries. Positives comprise 1,029 held-out source images, including synthetic imagery, and 810 novel Gaussian renders. Negatives comprise 1,029 training images and 810 renders from disjoint scene worlds within the nine scenes. Query composition varies by scene; Block A supplies 32.7% of positives. Methods share frozen queries, splits, calibration, and target scenes. Four scene scales are source-verified and five DA3-estimated [23] (Supplementary Table S8); calibration and interpretation are specified in Supplementary Sec. C.

Compared methods. Our CityLoC [26] reimplementation uses three fixed seeds and 64 pose proposals. QGVI-T retrieves 64 candidates and returns up to five distinct locations. Optional verification is reported as an ablation. QGVI-I uses the LoGS geometric initializer [8] without photometric refinement; its gate requires successful PnP and at least 40 distinct inlier query keypoints. The gate is calibrated on 490 Block A and SF validation queries and frozen across all nine test scenes (Sec. 3). LoGS and SplatHLoc [41] use their native maps; QGVI-I shares LoGS's RGB/SfM assets. Native LoGS accepts retrieval fallbacks; SplatHLoc requires four final inliers. We calibrate count and ratio gates for both LoGS pose variants, and a count gate for SplatHLoc, on the same validation set and correct-pose-retention objective, then freeze thresholds.

Model judgments. DeepSeek Flash supplies captions, descriptions, verification, and 176-query relevance labels, creating shared author–judge bias in that cohort. Historical 180- and 135-query scores use blind Qwen3-VL-8B-Instruct relevance judgments and separate DeepSeek image-only validity judgments. The completed historical Kimi/GPT reassessment appears in Supplementary Table S5. New controls use complete GPT judgments; partial Kimi assessments retain missing-label bounds. Judges and rubrics are kept separate (Supplementary Sec. D.2).

Common answer metrics. For each query's first answer, let S and W count correct and incorrect returns among N queries. Coverage is $(S + W)/N$, accepted-answer precision is $S/(S + W)$, correct-return rate is $Y = S/N$, and wrong-return rate is $E = W/N$. Following cost-based evaluation with rejection [14], we also report net query utility $U_\lambda = (S - \lambda W)/N$: a correct answer earns one unit, an incorrect answer costs λ, and abstention earns zero. Correctness means joint relevance/validity for text and an in-map

Table 1. **Shared answer accounting and net query utility.** First-answer results pooled over nine scenes; N counts query–seed observations for CityLoC and queries otherwise. C: coverage; P: accepted-answer precision; Y: correct-return rate; E: wrong-return rate. $100U_1$ rewards correct returns and penalizes wrong ones equally; abstention earns zero. Text correctness requires joint relevance and validity; RGB correctness requires an in-map pose within 1 m/2°. Text uses Qwen relevance and DeepSeek validity judgments; five RGB scenes have DA3-estimated scales. Panels are not comparable across modalities. Native rules and validation-calibrated RGB controls are distinguished; calibration shares a correct-pose-retention objective, not matched test coverage or risk. Bold marks best precision, correct-return rate, utility and wrong-return rate, including ties within each panel, including the optional verification ablation. ↑/↓ indicate higher/lower preferred; coverage and counts are descriptive.

Method	N	Coverage C (%)	Answer precision P (%) ↑	Correct returns Y (%) ↑	Wrong returns E (%) ↓	Net utility $100U_1$ ↑
Text queries (including optional verification)						
CityLoC w/o refinement	540	100.00	25.00	25.00	75.00	-50.00
CityLoC with refinement	540	80.19	32.56	26.11	54.07	-27.96
QGVI-T	180	100.00	58.33	58.33	41.67	16.67
QGVI-T + verification	180	75.00	**88.15**	**66.11**	**8.89**	**57.22**
RGB queries (50% out-of-map)						
LoGS, native	3,678	100.00	48.26	**48.26**	51.74	-3.48
LoGS + distinct gate	3,678	49.70	**97.10**	**48.26**	**1.44**	**46.82**
SplatHLoc, calibrated count	3,678	99.54	30.73	30.59	68.95	-38.36
QGVI-I	3,678	49.78	96.94	**48.26**	1.52	46.74

pose within 1 m/2° for RGB. Table 1 reports these frozen operating points with $U_1 = Y - E$: equal reward and penalty are an illustrative choice, not a calibrated deployment cost. Supplementary Sec. E gives the post-hoc cost sensitivity analysis. Text and RGB are evaluated separately; utility neither replaces pose error nor matches the methods' acceptance policies.

Text metrics. Top-K success (Hit@K) counts queries with a valid image satisfying all textual constraints among the first K outputs; empty lists fail. The 176-query cohort measures relevance only. Lower bounds count definite successes; upper bounds also count uncertain outcomes. These are label bounds, not confidence intervals or full-corpus recall. No-result rate counts empty returns, excluding runtime failures and missing judgments; it cannot certify that no matching view exists. Scene-macro scores weight scenes equally after CityLoC seed averaging; pooled scores weight queries.

RGB metrics. An in-map query depicts the target scene; an out-of-map query depicts another scene world. Translation error is camera-center distance; rotation error is the relative rotation angle. Pose success divides accepted poses within both 1 m/2° (strict) or 5 m/5° (loose) by *all in-map queries*, including rejected or missing poses as failures. Metre thresholds include five estimated scales. With in-map indicator y and acceptance indicator a,

$$\mathrm{FAR} = \frac{\sum(1-y)a}{\sum(1-y)}, \qquad \mathrm{FRR} = \frac{\sum y(1-a)}{\sum y}. \tag{5}$$

FAR and FRR use all out-of-map and in-map queries, respectively, and are independent of metre scale. Accepted wrong in-map poses are localization failures. Accepted-pose precision divides correct poses by all accepted outputs; conditional in-map precision excludes out-of-map outputs. Precision depends on the query mixture, balanced here.

4.2. Answer Quality and Localization Results

First-answer utility. In Table 1, QGVI-T yields 16.67 net reward units per 100 queries; optional verification yields 57.22 with lower coverage. On the separate RGB cohort, QGVI-I yields 46.74 and calibrated full LoGS 46.82, with equal correct-return rates but fewer wrong returns for LoGS (1.44% versus 1.52%). The SplatHLoc calibration retains its native operating point. Cross-modality utility values are not comparable.

Text-view retrieval. On the 180-query cohort, QGVI-T achieves 58.3% Top-1 and 78.3% Top-5 success, versus 26.1% and 34.6% for refined CityLoC (Supplementary Table S8(a)); no-result rates are 0.0% and 19.8%. All queries have joint judgments; macro and pooled rates agree after seed averaging. A one-query CityLoC repeatability check produced different refined poses and rankings despite fixed inputs and seed; no independent paired refinement effect has been measured.

Source poses and semantic judgments. On the same 180 descriptions, QGVI-T achieves 0.6% Top-1 source-pose success and 3.3% Pose Recall@5 at 5 m/5°; full-bank oracle coverage is 22.2%. Semantic retrieval scores therefore

do not establish source-camera accuracy. Supplementary Sec. D reports the pose audit and the separate, completed cross-model assessment.

RGB localization and acceptance. QGVI-I achieves 97.3% and 97.4% scene-macro pose success at 1 m/2° and 5 m/5° (Supplementary Table S8(b)). It accepts 13/1,839 out-of-map queries (0.707% pooled; 0.764% scene mean) and rejects 21/1,839 in-map queries (1.142% pooled; 0.765% scene mean). Accepted in-map poses include 43 strict and 40 loose localization failures. LoGS and SplatHLoc achieve 97.3% and 76.0% strict success, with 100.0% and 99.8% scene-mean false acceptance under native output rules. These native rates do not measure the benefit over calibrated gates. Every NYC and SF positive lies within 0.59 m of a training camera, with shared reconstruction dependencies; difficulty varies across scenes.

Table S16 isolates the acceptance gate by holding cached initializer outputs fixed. All 1,831 retained camera-to-world matrices are numerically unchanged. All 1,775 strict and 1,778 loose correct poses survive, preserving pooled success of 96.52% and 96.68%. The 21 rejected positives fail both thresholds. Strict accepted-pose precision rises from 48.26% to 96.94% and conditional in-map precision from 96.52% to 97.63%. Table 2 applies the same validation-retention objective to alternative rejection rules. Raw initializer inlier count accepts 24 distractors, versus 13 for distinct-keypoint support; both retain all strict-correct poses. Applying distinct support at the full LoGS pose accepts only ten distractors, with the same pose success and false-rejection rate as QGVI-I. Thus QGVI-I does not have the lowest false acceptance among these calibrated alternatives.

The paired photometric audit finds no strict or loose success/failure transitions, but all 3,678 pose matrices change. Over in-map queries, median translation/rotation errors increase from 0.012 m/0.023° to 0.028 m/0.085°. Equal threshold success therefore does not imply identical pose precision. Supplementary Sec. J reports all controls and continuous-error diagnostics.

4.3. Matched Text Retrieval Controls

Table 3 compares CLIP, BM25, MiniLM, and QGVI-T's rank fusion on the same 180 descriptions and rendered bank. Each method retrieves 64 candidates, forms a 20-view shortlist, and returns five distinct places, without online verification. GPT judges all 2,825 distinct query–image pairs. BM25 has higher confirmed Top-1/Top-5 success (62.8%/77.2%) than fusion (61.1%/73.9%); uncertainty intervals overlap. The experiment therefore does not establish a benefit from combining the two caption rankings. Full per-scene results and partial Kimi assessments appear in Supplementary Sec. H.

Table 2. **Rejection under a shared calibration objective.** Thresholds retain every validation-correct pose, then maximize rejection of validation distractors. Pooled test rates (%) use 1,839 queries per class; pose success is at 1 m/2°. LoGS includes photometric refinement. SplatHLoc retains its native requirement of at least four inliers. Bold marks the best displayed rates and ties.

Method / support threshold	Pose ↑	FAR ↓	FRR ↓
Initializer: count ≥ 40	**96.52**	1.305	1.033
Initializer: ratio ≥ 0.1	96.36	79.445	0.761
QGVI-I: distinct ≥ 40	**96.52**	0.707	1.142
LoGS: pairs ≥ 40	**96.52**	1.251	1.142
LoGS: distinct ≥ 40	**96.52**	**0.544**	1.142
SplatHLoc: native (≥ 4)	61.17	99.347	**0.272**

Table 3. **Matched text retrieval controls.** Same 180 queries, 30,522 rendered candidates, place selection, and five-result budget; no online verification. GPT judgments are complete. Ranges are label-uncertainty bounds (%), not confidence intervals. Bold marks the largest lower bound; overlapping ranges do not establish a ranking.

Retrieval	Top-1 ↑	Top-5 ↑
CLIP	39.4–54.4	62.2–80.6
BM25	**62.8**–78.9	**77.2**–92.2
MiniLM	56.7–73.9	70.6–90.0
QGVI-T (fusion)	61.1–77.8	73.9–91.7

Optional verification remains a separate operating-point ablation. Under the historical Qwen/DeepSeek rubric it raises Top-1 from 58.3% to 66.1% but lowers Top-5 from 78.3% to 72.8%, with 45/180 empty returns. The earlier 176-query controls and query-type breakdown are retained in Supplementary Tables S17 and S18.

4.4. Contribution of Rendered Grid Views

We compare an index of 18,570 original training images with an index that adds 30,522 Gaussian grid renders, keeping captions, retrieval settings, queries, and the five-result budget fixed. BM25 corpus statistics are recomputed for each combined bank. Original-image entries are judged using Gaussian renders at their stored cameras, matching the task of returning views within the supplied asset. All 1,329 distinct output pairs have GPT judgments; original-only predictions are reused unchanged.

Across 180 descriptions, confirmed Top-1/Top-5 success changes from 57.8%/73.3% to 60.0%/74.4% (Table 4). For the 106 descriptions authored from novel Gaussian renders, it rises from 61.3%/74.5% to 65.1%/77.4%. The 74 original-image descriptions show no Top-1 gain and a one-query Top-5 decrease. This pattern suggests a modest benefit on novel-view-derived descriptions. However, all paired label-uncertainty bounds include zero, and the source groups have different scene proportions. Same-pose render-

Table 4. **Adding grid views to original images.** Compare 18,570 original training images (O) with O plus 30,522 Gaussian grid renders (O+G). Each query searches its own scene. GPT-confirmed success (%); uncertain labels remain in the denominator. Bold marks higher confirmed success within each row and metric, including ties.

Description source	n	Top-1 ↑		Top-5 ↑	
		O	O+G	O	O+G
All descriptions	180	57.8	**60.0**	73.3	**74.4**
Original-image source	74	**52.7**	**52.7**	**71.6**	70.3
Novel-render source	106	61.3	**65.1**	74.5	**77.4**

Table 5. **Measured query latency.** Median/nearest-rank 95th percentile in seconds, pooled over all nine scenes. Text: 180 queries (540 observations per CityLoC arm); RGB: 36 fixed queries, three repeats. Empty outputs remain in the denominators. QGVI-T performs CPU retrieval without online VLM calls. Historical text timer boundaries differ; bold marks the fastest text latency and compares qualified RGB latencies.

Method	Query time (s) ↓ Median / p95
CityLoC w/o refinement	3.376 / 8.389
CityLoC with refinement	171.720 / 429.821
QGVI-T	**0.020 / 0.063**
LoGS	0.436 / 2.293
SplatHLoc	0.510 / 0.847
QGVI-I	**0.240 / 0.450**

ing, grid-only, density, and fixed-size replacement controls are reported in Supplementary Sec. I.

4.5. Query Latency

Table 5 reports recorded query measurements on nine scenes, retaining empty outputs. QGVI-T takes 0.020/0.063 s median/p95, including CPU encoding, retrieval, and place selection. Historical text timers differ and are not hardware-matched. CLIP/BM25/MiniLM controls record ranking-only time and are excluded.

RGB timing uses four queries per scene, three resident repeats, and an RTX PRO 6000 Blackwell GPU. Timers include decoding, preprocessing, localization, acceptance, and device synchronization, excluding map/model loading. Additional calibrated gates have no separate timing measurement. Supplementary Sec. D.3 gives per-scene times, timer boundaries, and repeatability checks.

5. Discussion and Limitations

Pose preservation and acceptance. QGVI-I preserves accepted pose coordinates by construction, but retention of every correct initializer pose is empirical. Accurate poses with fewer than 40 distinct inliers could be rejected; accepted geometric support also does not certify accuracy. The rule provides no distribution-free risk bound. Full LoGS with the same distinct-support criterion rejects three more distractors at unchanged threshold success, while changing continuous pose errors and incurring photometric refinement. The fixed-initializer gate therefore offers a different cost/accuracy trade-off; it does not establish the general ineffectiveness of photometric refinement.

Text success and abstention. Optional verification improves Top-1 but reduces Top-5 success. Its 25.0% no-result rate does not measure correct rejection: failure to verify a shortlist cannot establish absence of a matching view. RGB distractors come from other benchmark scene worlds; their rejection does not establish robustness to arbitrary appearance, sensor, or content changes.

Retrieval and view coverage. BM25-only retrieval remains competitive with rank fusion under matched controls. Adding rendered grid views produces modest confirmed gains on novel-view-derived descriptions, but unresolved labels prevent a conclusive paired improvement claim. Candidate count, camera-domain coverage, and the scene composition of each description group limit causal interpretation.

Evaluation scope. Model-judge agreement measures consistency rather than human correctness. DA3 supplies a fixed scale conversion where source metric scales are unavailable, not surveyed ground truth. Reconstruction reuse and scale calibration are specified in Supplementary Sec. C.

6. Conclusion

Queryable Gaussian View Index queries supplied Gaussian scene assets through complementary text-view retrieval and selective camera localization routes. QGVI-T retrieves distinct places from structured descriptions of posed Gaussian views, without online verification. QGVI-I gates LoGS-based geometric estimates using distinct inlier query keypoints while preserving accepted poses. On the nine-scene development benchmark, QGVI-T reaches 78.3% Top-5 success; the RGB gate rejects most out-of-map queries while retaining every correct initializer pose. Matched controls find competitive BM25 retrieval and small, uncertain gains from view-bank expansion. A calibrated full-LoGS gate slightly improves rejection at the same pose-success thresholds. Shared coverage, precision, and correct-return accounting retains the distinct semantic and geometric definitions of correctness.

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A. Supplementary Material

A.1. Historical Gaussian-Depth Pipeline

The historical Gaussian-depth pipeline (v3) localizes images using Gaussian-rendered RGB-D references and geometric verification. It differs from QGVI-I in retrieval, local matching, 3D geometry, and verification. Table S1 compares these complete designs; Table S16 isolates the QGVI-I acceptance gate.

The historical pipeline uses SALAD [16] retrieval with four diverse starting views, RootSIFT [2] correspondences based on SIFT [25], geometric pose estimation, and verification against a newly rendered Gaussian RGB-D view. The query is resized to a maximum side of 512 pixels before SALAD's standard 322-pixel input transform. The reference bank, 512-pixel local features, solver, gates, and render budgets remain fixed. CLIP embeddings are retained for diagnostics and do not determine geometric acceptance. This variant uses the original Horizon-GS [17] checkpoints and frozen native reference banks.

Each reference is back-projected with its own intrinsics. Pose estimation and verification renders use the query intrinsics, so retrieval does not require the reference and query cameras to have matching calibration.

The historical pipeline achieves 81.1% strict scene-macro pose success, accepts no out-of-map queries, and rejects 381 in-map queries (20.7% pooled; 16.0% scene mean). It also accepts 45 inaccurate poses; these localization failures are distinct from false rejects. QGVI-I has 97.3% strict scene-macro pose success, 0.707% pooled false acceptance, and 1.142% pooled false rejection on the same queries. The historical pipeline rejects more in-map queries while accepting no out-of-map query in this cohort. Neither result establishes a universal rejection guarantee.

A.2. Representation and Initialization Diagnostics

Map representation. With the original maps, the historical pipeline accepts 1,458 in-map poses, of which 1,413 satisfy 1 m/2°. Pose success is 76.8% when pooled over all 1,839 in-map queries and 81.1% when scenes receive equal weight; neither denominator excludes rejections. On identical query images and reference camera poses, the earlier common-SH3 adaptation achieves 19/139 strict successes in Block Small and 9/602 in Block A, compared with 88/139 and 365/602 for the original-map system. Reference renders, depth, retrieval encoder and selections, and camera adapter change together, preventing attribution to Gaussian map quality alone. The original maps follow the benchmark configuration, but the transition was evaluated after inspecting earlier outcomes. This is therefore a development comparison; the common-SH3 scores remain separate from Table S1.

Query composition. The earlier 96.0% result uses a separate set of 200 rendered Block A queries, equally split between aerial and street views, inside a supported 256-patch atlas with 24,576 references. That cohort is distinct from the 602 Block A positives in the nine-scene benchmark. Using the same atlas on these 602 queries yields 97 strict successes (16.1%), compared with 365 (60.6%) for the historical SALAD-initialized system using the original representative bank. This comparison changes the initializer as well as reference coverage. The 96.0% estimate therefore characterizes the supported-atlas setting; it does not describe performance on the nine-scene pool.

A source-image comparison identifies 374 queries shared with the earlier Block A benchmark. Source pixels, camera rotations, metre-converted translations and resized intrinsics agree for every matched image. The earlier CLIP-initialized native-v3 run has 113 strict successes (30.2%); the historical SALAD-initialized system has 231 (61.8%), versus 306 (81.8%) for the archived Gaussian-only SALAD control. Cohort membership and pose labels alone therefore do not explain the remaining gap. The control changes retrieval, reference views, local feature resolution to 960 pixels, and verification; it is excluded from the main method comparison.

Camera calibration. An earlier same-intrinsics restriction excluded 16 of Park's 228 in-map queries and limited some Road queries to 12 of 2,669 references. Removing that restriction and rendering at query calibration increases strict success from 80 to 109 queries in Park and from 47 to 75 in Road. Colosseum and Elvenruin each lose one success; the other scenes are unchanged. The restricted adapter rejected 1,518 negatives without matching, whereas the corrected CLIP-initialized system matches all 1,839. With CLIP, Block A retains 172 strict successes; of its 2,408 positive-query initial hypotheses, 1,324 have insufficient matches and 590 fail RANSAC. These counts locate the dominant loss at initialization; they do not isolate retrieval from appearance or reference coverage.

Initializer selection. A paired comparison on all 988 validation queries holds the reference bank, local geometry, budget of four starting views, and acceptance gates fixed. Replacing CLIP ranking with SALAD improves strict scene-macro pose success from 62.6% to 80.0%, with zero accepts among 494 negatives in each arm. The selection rule requires higher strict success, no increase in either scene-macro false acceptance or total negative accepts, and no additional execution failures. The SALAD recipe was fixed before the full test run and was not retuned on its outcomes. On the complete test cohort, strict success increases from 62.7% to 81.1%, including 172 to 365 strict successes in Block A. Both query pools retain historical development

Table S1. **Historical Gaussian-depth pipeline and QGVI-I.** Both use the same nine-scene development queries and fixed scale factors. Pose success uses all in-map queries; false acceptance uses all out-of-map queries; false rejection uses all in-map queries. Rates are percentages; scene means weight scenes equally. † DA3-estimated scale. This comparison changes several components jointly and does not isolate the acceptance gate. Bold marks best values within each scene and metric, including ties.

Method	Metric (%)	Block Small	Block A	NYC	SF	Citysample	Colosseum	Elvenruin	Park	Road	Scene mean
In-map queries	n	139	602	180	174	142	126	118	228	130	—
Out-of-map queries	n	139	602	180	174	142	126	118	228	130	—
Scale	m / native unit	25	25	1	1	24.777†	12.287†	14.135†	1.040†	1.339†	—
Gaussian-depth pipeline (v3)	Pose 1 m/2° ↑	63.3	60.6	**100.0**	**100.0**	88.0	64.3	94.9	82.9	76.2	81.1
	Pose 5 m/5° ↑	64.0	62.0	**100.0**	**100.0**	95.8	64.3	97.5	84.2	78.5	82.9
	False accept ↓	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
	False reject ↓	36.0	37.7	**0.0**	**0.0**	4.2	29.4	2.5	14.0	20.0	16.0
QGVI-I	Pose 1 m/2° ↑	**90.6**	**93.4**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.3**
	Pose 5 m/5° ↑	**92.1**	**93.5**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.4**
	False accept ↓	**0.000**	0.332	1.111	4.023	1.408	**0.000**	**0.000**	**0.000**	**0.000**	0.764
	False reject ↓	**3.597**	**2.492**	**0.000**	**0.000**	**0.000**	**0.794**	**0.000**	**0.000**	**0.000**	**0.765**

exposure.

Unresolved geometric conventions. The source audit identifies an unexpressed half-pixel conversion between Gaussian raster coordinates and OpenCV feature-depth lifting. A conditional CPU consistency replay changes initial poses without changing gates; the source-image pixel convention and final localization effect remain unresolved. No calibration correction is applied to the historical pipeline in Table S1. Repeated SIFT orientations can also count one physical landmark more than once in correspondence support. These limitations remain separate from the measured initializer comparison.

A.3. Reference-View Coverage

Orientation coverage is evaluated in the historical CLIP-initialized common-SH3 adaptation using three reference banks: the baseline grid, an equal-size bank replacing selected views with upward-facing views at the same grid positions, and a bank containing all additional upward-facing views. The equal-size comparison holds the reference count fixed and retains every query. On Block Small, both modified banks add three successful queries at each pose threshold: success at 1 m/2° increases from 19/139 (13.7%) to 22/139 (15.8%). Full expansion adds no further successes over equal-size replacement. Block A remains at 9/602 (1.5%) for both thresholds under all three banks. None of the six runs accepts an out-of-map distractor. These results concern the two exposed development scenes and the selected distractor pools.

A.4. Historical Retrieval Measurements

For the historical pipeline, retrieval timing includes reading, decoding, and resizing the query, SALAD and retained CLIP encoding, and SALAD ranking with selection of four starting views. GPU synchronization follows encoding. Model/index loading, pose search, render verification, and output writing are excluded; diagnostic descriptor writes between timed stages are also excluded. These measurements characterize the historical initialization stages, not optimized production latency or QGVI-I. Table S3 also preserves CLIP-based timings from separate camera-support configurations and the 200-query Block A atlas. Their cohorts and configurations remain distinct.

B. Implementation Details

Posed view bank. The nine-scene text bank contains 30,522 rendered views at 2,330 distinct place identifiers. Camera centers form a deterministic lattice near the training cameras; heights are interpolated from nearby training cameras, and intrinsics are borrowed from training views. Each place identifier combines the scene, camera domain, and camera-center hash. Different orientations at one center therefore share an identifier; a place is not a learned semantic landmark or a spatial-radius cluster. This construction extends the candidate observations beyond captured views. Its effect on retrieval accuracy is evaluated using original-only and original-plus-grid controls in Sec. 4.4; density and replacement studies are reported in Supplementary Sec. I.

QGVI-T and optional verification. Each structured caption contains a summary and fields for objects, attributes, relations, visible text, and uncertainty. Retrieval indexes the descriptive fields and excludes uncertainty to avoid indexing that field as positive retrieval evidence. BM25 uses $k_1 = 1.2$ and $b = 0.75$. Sentence scores are pooled over the complete index before the view ranking is truncated; view identifiers break fusion-score ties. Among the 64 candidates, the first occurrence of each place identifier precedes

Table S2. **Auxiliary common-map reference-bank ablation for RGB localization.** These six historical CLIP-initialized common-SH3 runs share the original query IDs and one fixed map; they are separate from the historical native-map comparison in Table S1. Denominators are 139 in-map and 139 out-of-map queries for Block Small, and 602 of each for Block A. Query-time quantiles cover all queries, including rejected ones, from encoding through retrieval, pose search, verification when reached, and result transfer; model/cache startup and queue waiting are excluded. Quantiles use linear interpolation between ordered query times. Within each scene, bold marks higher pose success and lower false acceptance or time, including ties. Median and 95th-percentile times are compared separately.

Scene	Reference bank	Views	Pose success ↑ 1 m/2° (%)	Pose success ↑ 5 m/5° (%)	Out-of-map accept (%) ↓	Query time (s) ↓ Median / 95th percentile
Block Small	Baseline	1,886	13.7	14.4	**0.0**	0.117 / 0.588
	Equal-size replacement	1,886	**15.8**	**16.5**	**0.0**	**0.109 / 0.587**
	Full expansion	1,945	**15.8**	**16.5**	**0.0**	**0.109** / 0.591
Block A	Baseline	13,094	**1.5**	**1.5**	**0.0**	0.125 / 0.221
	Equal-size replacement	13,094	**1.5**	**1.5**	**0.0**	**0.124 / 0.209**
	Full expansion	14,347	**1.5**	**1.5**	**0.0**	0.126 / 0.214

Table S3. **Historical RGB retrieval-stage latency.** Times are milliseconds; quantiles use linear interpolation and include rejected queries. (a) Historical SALAD-initialized geometric-v3, including retained CLIP encoding. (b) Separate CLIP-based configurations and cohorts. Timings are descriptive and are not ranked across scenes or cohorts.

(a) Nine-scene historical geometric-v3

Scene	Queries	Median (ms) ↓	95th percentile (ms) ↓
Block Small	278	53.184	72.137
Block A	1,204	101.922	122.144
NYC	360	50.939	73.183
SF	348	59.176	86.112
Citysample	284	54.040	59.646
Colosseum	252	44.233	59.522
Elvenruin	236	46.095	62.974
Park	456	59.687	86.181
Road	260	59.922	79.409
Pooled	3,678	61.076	120.157

(b) Other historical configurations

Benchmark	Initializer	Queries	Median (ms) ↓	95th percentile (ms) ↓
Nine-scene RGB, cross-camera	CLIP (historical)	3,678	34.046	75.635
Nine-scene RGB, camera-filtered	CLIP (historical)	3,678	30.529	89.236
Block A supported atlas	CLIP (historical)	200	43.817	47.198

repeated occurrences, with relative order preserved within both groups. The first 20 views form the shortlist from which QGVI-T returns five distinct places. Only QGVI-T + verification applies visual verification to that shortlist. Both offline captioning and optional verification use DeepSeek Flash at temperature zero with a maximum image side of 1,280 pixels. Verification receives the image without its caption and produces a relevance score used only for ranking.

QGVI-I local features. The LoGS geometric initializer uses at most 4,096 SuperPoint keypoints, a 1,024-pixel maximum image side, non-maximum suppression radius three, and outdoor SuperGlue weights with five Sinkhorn iterations.

Acceptance calibration. The calibration set contains 245 in-map and 245 out-of-map queries from Block A and SF. A predefined grid tests 343 combinations of minimum distinct inliers, inlier fraction, and image coverage. Eligible policies retain every validation pose correct at either 1 m/2° or 5 m/5°. Selection first maximizes out-of-map rejections, then retained in-map queries; remaining ties favor fewer active gates and smaller thresholds. The selected rule requires successful PnP and at least 40 distinct inlier query keypoints within 12 pixels at the saved pose. Fraction and coverage

thresholds are zero. This policy is fixed across all nine test scenes.

C. Additional Experimental Protocol and Diagnostics

Source split and text cohorts. The localization source split contains 18,570 training and 1,309 evaluation images with no exact file overlap. Some evaluation observations contributed to reconstruction; the evaluation therefore does not test generalization to unseen reconstruction observations. In the 180-query text cohort, source-image review labels 160 descriptions fully supported, 14 partially supported, and six unsupported. All remain in the denominator. A description unsupported by its source image may still match another view. Development exposure includes 96 previously used source images and one repeated normalized description. The reported cohorts are development evaluations; no independent final evaluation has been conducted. The 180- and 135-query text cohorts use RGB test-source images; the 176-query cohort uses RGB validation-source images. The 135 descriptions are synthetic; the other two cohorts are DeepSeek-authored. Within each comparison, methods share query IDs, source splits, and target scenes. The overall text-ablation and breakdown rows pool the 176 query IDs.

RGB cohorts and cached evaluation. The 1,839 test positives comprise 480 aerial and 1,359 street views. Block A contributes 501 source images and 101 renders, versus 13 and 113 in Colosseum. The separate 988-query validation pool contains 280 source-image positives, 214 rendered positives, and 494 negatives. Out-of-map queries have no pose error and enter false-acceptance evaluation, not pose-success denominators. An always-accepting method has 100% false acceptance regardless of in-map accuracy. Initialization failures remain in the reported pose-success denominators. QGVI-I evaluation reuses frozen initializer outputs and correspondences; fresh runs on four queries reproduce their cached outputs. The GSplatLoc appearance diagnostic is excluded from scored comparisons.

The 13 QGVI-I false accepts occur in Block A (2), NYC (2), SF (7), and Citysample (2). False rejects occur in Block Small (5), Block A (15), and Colosseum (1). Map-training coverage also differs: the released LoGS schedule loads 3,100 of Park's 6,194 allowed cameras over 30,000 iterations; the number of distinct cameras used in optimization is unrecorded. All nine scenes are reported.

Scene-scale verification. The source-verified scales are 25 metres per model unit for Block Small and Block A, and one for NYC and SF. The latter are checked against 351 paired camera shifts with labelled 0.3 m height differences and 774 source-to-checkpoint and checkpoint-to-benchmark camera matches per scene. The other five scene scales use a calibrated ratio of DA3 raw depth to Gaussian-rendered expected camera-Z depth on frozen training street views. A conversion constant is fitted using 13 Block Small views and its verified scale, then applied unchanged to the median depth ratio in each target scene. Every method uses the same frozen scene-wide factor; it is not fitted to localization outcomes. Block A, held out from scale calibration, has a scale error of 3.9%. Calibration cameras retain their native coordinates, so scale transfer does not require map rescaling. This procedure provides a shared metric conversion for comparison, while the five DA3 estimates remain unverified by independent physical measurements.

Text retrieval diagnostics. On the 135-query cohort, the optional verification variant produces three Top-5-only gains and seven losses relative to caption retrieval, a net loss of four. Six losses return no result; the seventh ranks its relevant view sixth. All nine successful caption-retrieved views from these seven queries were in the 20-view shortlist. These observations do not support a tuning claim. For the 176-query shortlists, caption and CLIP success-count bounds are 152–176 and 141–175; these are not full-bank recall estimates. Returned-view validity is judged for the 135- and 180-query cohorts but unverified for the 176-query cohort. On the 180-query cohort, CityLoC refinement raises seed-averaged Top-5 success from 30.7% to 34.6%, with 19.8% no result.

Historical query timing boundaries. The earlier 176-query efficiency run uses PyTorch 2.9.1+cu128. CityLoC timings include serialization and GPU synchronization, exclude model/map startup, and use all queries with nearest-rank 95th percentiles. Peak PyTorch allocated memory includes resident models/maps but excludes driver and custom non-PyTorch allocations. The separate 135-query pilot uses 135 IDs across three seeds, including only one ID each in NYC and SF; per-scene median times are 2.5–7.3 s without refinement and 35.6–287.6 s with it. Returning no result does not establish correct rejection.

That QGVI-T + verification timing pass uses 3,562 HTTP attempts across all 176 queries and retries (20.24 per query); two query IDs end in API-response failure. The execution history contains four failed executions, including earlier retries; this count differs from the two failed query IDs. Failed executions enter neither latency quantiles nor retrieval no-result counts. Cached-response diagnostics are excluded from online timing. Provider GPU usage and memory are unknown.

D. Additional Evaluation Protocols

The following evaluations separate recovery of the description's source camera, semantic satisfaction, and query latency. The 180-query development cohort and method outputs remain fixed.

D.1. Text Source-Camera Recovery

Each description is linked to the camera that produced its authoring image. For a predicted camera-to-world pose $(\hat{R}, \hat{c})$ and source pose $(R^\star, c^\star)$, translation error is $s\|\hat{c} - c^\star\|_2$, where s is the frozen scene scale; rotation error is the angle of $\hat{R}^\top R^\star$. Stored retrieval poses can be evaluated directly. These errors measure source-camera recovery: another view may satisfy the description without being close to its source camera.

Top-1 pose success requires both translation and rotation errors to meet the specified threshold. Pose Recall@5 requires at least one of the first five outputs to meet both thresholds; the same output must satisfy both. Empty returns fail, and shorter lists receive no additional candidates. Missing query records or invalid pose matrices invalidate an evaluation rather than becoming successful abstentions. All methods use the same query IDs, fixed ranking, and output budget. CityLoC's three seeds are averaged within each query before scene averaging.

We report the nine-scene mean and a separate mean over the four scenes with source-verified metre scales. The remaining five use the fixed DA3 estimates disclosed in Table S8. Top-1 error distributions are conditional on a returned valid pose and must be accompanied by their denominator and return rate. An auxiliary full-bank oracle tests whether any stored reference pose can satisfy the thresholds; it measures the discrete bank's coverage, not the retrieval method's accuracy. The bank covers 0/180 source cameras at 1 m/2° and 40/180 at 5 m/5°. For 171 descriptions, no bank orientation is within 2° of the source, irrespective of position. Independent checks trace all source poses to the original held-out or novel-view records and reproduce the pose scores with a second rotation implementation. This coverage bounds QGVI-T's discrete retrieval, not CityLoC's continuous pose proposals. At the looser threshold, QGVI-T retrieves 6 of the 40 queries covered by the bank within its five outputs; the optional verification variant retrieves 8.

D.2. Blinded Cross-Model Semantic Assessment

The historical CityLoC/QGVI-T comparison in Table 1 and the earlier 180- and 135-query tables retain completed Qwen/DeepSeek judgments. Qwen3-VL-8B-Instruct (revision `0c351dd0`) assigns one of four relevance labels: fully relevant, partially relevant, irrelevant, or uncertain. Separate DeepSeek image-only judgments assign render validity. Joint success requires full relevance and a valid image; uncertain labels produce lower and upper bounds. The completed Kimi/GPT assessment is reported separately in Table S5; its labels do not replace or augment the main table's historical judgments.

The completed reassessment pools fixed Top-5 outputs from CityLoC with and without refinement and QGVI-T with and without verification, across all seeds on the 180-query cohort: 6,150 output occurrences yield 5,808 distinct query–pixel pairs. Identical pairs are assessed once per reviewer and their labels reused for every occurrence. Each reviewer receives only the description and rendered image; method, rank, camera pose, stored captions, and earlier judgments are withheld.

Reviewer A uses Kimi (requested model `kimi-for-coding`, Kimi Code CLI 2.2.0, low thinking effort); Reviewer B uses GPT-6.1 (requested model `gpt-6.1-sol`, Codex CLI 0.159.2, low reasoning effort). These are requested model aliases; backend version identities are not independently verified. They are called independently with the same rubric: relevance is supported, contradicted, or uncertain, and validity is valid, invalid, or uncertain. Their joint success criterion requires supported relevance and valid imagery. Their judgments and disagreements are retained separately. Failed or invalid responses were retried without replacing valid judgments. Both reviewers completed all 5,808 pairs (11,616 judgments); uncertain judgments remain unresolved. This three-way relevance rubric differs from Qwen's historical four-way rubric: a partially relevant judgment may correspond to contradiction or uncertainty. Comparing full support across the two rubrics therefore does not imply exact category agreement.

Top-1 and Top-5 success include all queries, including empty returns; View-level precision pools all returned views, whereas the primary accepted-answer precision in Sec. E uses only the first answer per query. No-result rate measures empty outputs without establishing whether a description is unanswerable. Full-corpus recall is unavailable because the complete relevant-view set is unknown. These are model-generated labels, and inter-model agreement alone does not establish their correctness.

Table S5 reports separate Top-1 and Top-5 scores and first-answer accounting for these four historical text arms. For A returned answers, S confirmed successes, W confirmed wrong answers, and u unresolved answers, precision lies in $[S/A, (S+u)/A]$ and wrong-return rate in $[W/N, (W+u)/N]$. The utility interval is $[(S-W-u)/N, (S+u-W)/N]$; abstention earns zero. Both judges give QGVI-T higher Top-5 lower bounds than its verified variant; their Top-1 lower-bound ordering differs. Overlapping intervals limit ranking claims. These are repeated assessments of the same development queries, not an independent test set.

Table S4. **Text source-camera recovery.** Same development descriptions; scene-macro percentages after seed averaging. P1 is Top-1 pose success and R5 is Pose Recall@5. Error columns average conditional scene medians, with coverage reported alongside the completed results. CityLoC first averages errors over seeds returning a valid Top-1 pose within each query. The four-scene subset uses source-verified metre scales; five other scenes retain DA3 estimates. All scenes contribute conditional errors. In row order, error distributions contain 180, 160, 180, and 135 queries (80, 72, 80, and 63 in the four-scene subset); pose coverage averages seed-level returns. Bold marks highest success and lowest error within each cohort; coverage is descriptive.

Method	P1 ↑ 1 m/2°	R5 ↑ 1 m/2°	P1 ↑ 5 m/5°	R5 ↑ 5 m/5°	Mean scene median ↓ (m/°)	Pose coverage (%)
All nine scenes, 180 queries						
CityLoC w/o refinement	**0.00**	**0.00**	0.19	0.19	86.30 / 69.99	100.00
CityLoC with refinement	**0.00**	**0.00**	0.37	0.56	89.16 / 64.98	80.19
QGVI-T	**0.00**	**0.00**	**0.56**	3.33	64.66 / **51.64**	100.00
QGVI-T + verification	**0.00**	**0.00**	**0.56**	**4.44**	**47.08** / 55.26	75.00
Four scenes with source-verified scales, 80 queries						
CityLoC w/o refinement	**0.00**	**0.00**	0.42	0.42	125.34 / 68.42	100.00
CityLoC with refinement	**0.00**	**0.00**	0.83	0.83	132.09 / 62.06	80.42
QGVI-T	**0.00**	**0.00**	**1.25**	6.25	90.57 / **51.12**	100.00
QGVI-T + verification	**0.00**	**0.00**	**1.25**	**7.50**	**60.33** / 51.75	78.75

Table S5. **Separate Kimi and GPT assessments of all text methods.** All 180 development queries from nine scenes; each CityLoC variant has three seeds, giving $N = 540$ query–seed observations. Coverage and all quality rates are percentages. Top-1 success is correct-return rate Y; precision P conditions on returned answers, while wrong-return rate E divides by all queries. Net utility $100U_1$ uses per-query reward $+1$ for a correct first answer, -1 for a wrong answer, and zero for abstention. Ranges are unresolved-label bounds, not confidence intervals. Bold marks the best conservative bound within each reviewer: lower for success, precision and utility, upper for wrong returns. Overlap limits ranking claims. Native acceptance policies are retained; the main table separately uses historical Qwen relevance and DeepSeek validity labels.

(a) Coverage, retrieval success, and first-answer utility

Method	N	Coverage C	Top-1 success Y ↑		Top-5 success ↑		Net utility $100U_1$ ↑	
			Kimi	GPT	Kimi	GPT	Kimi	GPT
CityLoC w/o refinement	540	100.0	22.2–61.1	26.7–66.5	37.8–84.6	40.6–82.0	[−55.6, 22.2]	[−46.7, 33.0]
CityLoC with refinement	540	80.2	25.6–47.0	29.1–49.4	37.4–64.1	39.6–63.0	[−29.1, 13.9]	[−22.0, 18.7]
QGVI-T	180	100.0	52.2–73.9	**61.1**–77.8	**72.8**–93.3	**73.9**–91.7	[4.4, 47.8]	[22.2, 55.6]
QGVI-T + verification	180	75.0	**56.7**–68.9	60.6–68.9	67.2–74.4	68.3–72.8	[**38.3**, 62.8]	[**46.1**, 62.8]

(b) First-answer precision and wrong-return rate

Method	Answer precision P ↑		Wrong returns E ↓	
	Kimi	GPT	Kimi	GPT
CityLoC w/o refinement	22.2–61.1	26.7–66.5	38.9–77.8	33.5–73.3
CityLoC with refinement	31.9–58.7	36.3–61.7	33.1–54.6	30.7–51.1
QGVI-T	52.2–73.9	61.1–77.8	26.1–47.8	22.2–38.9
QGVI-T + verification	**75.6**–91.9	**80.7**–91.9	6.1–**18.3**	6.1–**14.4**

D.3. Measured Query Latency

We report measured query latency for seven historically timed variants on nine scenes. New matched retrieval controls have ranking-only timings; calibrated baseline gates have no separate query timing. Timers exclude model and reference loading. Cached-response replay is not used as a fresh API query measurement.

Text queries. CityLoC with/without refinement and QGVI-T with/without verification use the same 180 descriptions. CityLoC has 540 query–seed observations per arm. Without refinement, its median/p95 is 3.376/8.389 s; with refinement, 171.720/429.821 s, with 107 empty outputs. Both arms render diffusion proposals for CLIP selection and therefore require the inherited Gaussian map. Their timers include candidate-image serialization and

hashing, but exclude model/map loading. QGVI-T takes 0.020/0.063 s, including diversity selection. QGVI-T + verification takes 15.465/17.067 s; all 180 queries complete, 45 return no result, and the timer includes 3,614 HTTP attempts, network round trips, and retries. Its output lists differ from the frozen accuracy predictions on 73 queries, so timing-pass outcomes are not substituted for accuracy results. QGVI-T runs on the CPU; its optional verification variant adds a remote service. The historical text runs were recorded on different dates without a matched hardware protocol; the remote service hardware is unknown.

Table S6 reports each scene separately. Each CityLoC cell pools 20 queries over three training seeds; each QGVI-T variant has 20 queries per cell. Quantiles are calculated from the individual query times, not averaged across seed or scene summaries.

Fresh RGB timing. All three RGB methods use the same 36 query IDs, selected by a fixed hash of public IDs before timing, and three resident repetitions per query. Images, calibration, models, and reference assets retain their frozen protocols. QGVI-I requires fresh features, retrieved references, matches, poses, and gate decisions to reproduce its saved implementation outputs. LoGS checks exact initializer outputs and canonical camera, image, mask, and optimizer inputs before executing the unchanged native refinement. SplatHLoc checks public input pixels and its unchanged native configuration. For the two refined baselines, differences from historical poses are retained as repeatability diagnostics.

LoGS's initial numerical check also failed when rerunning its frozen native implementation. In an eight-query diagnostic over Block Small and NYC, 7/48 direct native calls exceeded the previous saved-pose tolerance. Identical visible forward outputs and losses produced different pose gradients. Subsequent timing therefore qualifies native source and exact external-input conformance; numerical variation is reported separately, without increasing the tolerance or selecting closer runs. This does not establish that the variation is harmless. The failed checks and fixed accuracy outputs are preserved.

For LoGS, SplatHLoc, and QGVI-I, timing starts before image decoding and ends after the pose and acceptance decision with device synchronization. Model/reference loading and qualification checks are outside the query timer. The timing subset does not estimate full-test accuracy or represent an unseen test. All local RGB geometry runs use an NVIDIA RTX PRO 6000 Blackwell Workstation Edition. LoGS retains input references and clones eight initial optimizer scalars inside the timer; input hashing and canonical-input checks follow it. PyTorch uses two CPU threads; native OpenCV threading is retained (32 threads recorded for SplatHLoc), so this is not a two-core host limit. Quantiles are computed from pooled raw observations, with nearest-rank 95th percentiles. With twelve observations per scene, this percentile is the observed maximum; this small timing sample does not establish a stable population tail.

E. Shared Answer Accounting and Net Query Utility

Both query routes return an answer whose correctness can be evaluated, but their correctness predicates differ. For text, the first returned view must be usable and satisfy every required semantic constraint. For RGB, the query must be in-map and the returned pose must satisfy both geometric thresholds. An accepted out-of-map image or an inaccurate in-map pose is incorrect. This shared accounting follows selective prediction [15]; it does not equate semantic satisfaction with geometric accuracy.

For N queries, let $a_i = 1$ when query i receives a first answer and $z_i = 1$ when that answer is correct. Set $a_i z_i = 0$ for an empty return. With $A = \sum_i a_i$, $S = \sum_i a_i z_i$, and $W = A - S$, define

$$C = \frac{A}{N}, \qquad P = \frac{S}{A}, \qquad Y = \frac{S}{N} = CP, \qquad E = \frac{W}{N}. \tag{6}$$

Coverage C measures how often the method answers; accepted-answer precision P measures how often those answers are correct; correct-return rate Y measures how often a query receives a correct answer. Wrong-return rate E counts incorrect answers over all queries. Selective risk is $1 - P$. If $A = 0$, precision and selective risk are undefined, while $C = Y = E = 0$. The identity uses fractions and pooled counts; it need not hold after independently averaging scene-level C and P.

Explicit costs for wrong answers. Cost-based evaluation makes the relative consequences of answering and abstaining explicit [14]. We assign reward $+1$ to a correct first answer, reward $-\lambda$ to an incorrect first answer, and zero to abstention:

$$U_\lambda = \frac{S - \lambda W}{N} = Y - \lambda E, \qquad \lambda \geq 0. \tag{7}$$

An all-reject system scores zero; a system returning only wrong answers scores negatively for $\lambda > 0$. The identities $U_0 = Y$ and $U_1 = 2Y - C$ show that utility adds an explicit preference over the existing counts, rather than additional evidence. We report $100U_\lambda$ in net reward units per 100 queries, not as an accuracy percentage. Values can fall below -100 when $\lambda > 1$.

The cost grid $\lambda \in \{0, 0.5, 1, 2, 5\}$ is a descriptive sensitivity analysis specified after the recorded evaluations. It was not used to select methods or calibrate thresholds. Equal reward and penalty ($\lambda = 1$) provide an interpretable

Table S6. **Text query latency by scene (seconds; ↓).** Median/nearest-rank p95 from the recorded query runs. CityLoC has 60 observations per cell and each QGVI-T variant has 20. Empty outputs remain included. The verified text route includes remote verification and retries.

Scene	CityLoC w/o refinement	CityLoC with refinement	QGVI-T	QGVI-T + verification
Block Small	4.776 / 5.915	145.724 / 609.644	0.021 / 0.029	15.457 / 16.844
Block A	8.244 / 10.336	182.853 / 700.906	0.062 / 0.086	15.648 / 17.970
NYC	3.297 / 3.693	198.926 / 327.830	0.018 / 0.022	15.462 / 16.845
SF	2.717 / 3.660	177.899 / 347.767	0.020 / 0.024	15.334 / 16.155
Citysample	3.113 / 3.772	85.713 / 239.108	0.021 / 0.024	15.380 / 16.164
Colosseum	2.559 / 3.081	170.950 / 226.452	0.012 / 0.015	15.836 / 16.915
Elvenruin	3.411 / 3.974	115.690 / 429.821	0.011 / 0.014	15.293 / 16.075
Park	3.527 / 4.066	180.399 / 332.218	0.029 / 0.037	16.092 / 18.325
Road	3.453 / 4.046	151.034 / 257.865	0.019 / 0.027	15.077 / 17.638

Table S7. **RGB latency by scene (seconds; ↓).** Median/nearest-rank p95 of twelve resident observations per method and scene. Bold marks the lowest displayed median and p95 across three qualified methods, including ties.

Scene	LoGS	SplatHLoc	QGVI-I
Block Small	0.487 / 0.789	0.529 / 0.630	**0.286 / 0.342**
Block A	0.425 / 0.722	0.527 / **0.544**	**0.329** / 0.647
NYC	1.065 / 2.732	0.399 / 0.877	**0.193 / 0.246**
SF	0.365 / 0.776	**0.107** / 0.857	0.232 / **0.300**
Citysample	0.343 / 1.424	0.543 / 0.649	**0.198 / 0.262**
Colosseum	0.318 / 0.366	**0.118** / 0.613	0.227 / **0.287**
Elvenruin	0.564 / 2.305	0.530 / 0.658	**0.222 / 0.334**
Park	0.508 / 0.979	0.385 / 0.735	**0.229 / 0.307**
Road	0.459 / 0.554	**0.108** / 0.588	0.259 / **0.453**

example; no deployment cost has been established. This standard utility construction is an evaluation choice, not a methodological contribution.

Table 1 reconstructs these operating points from frozen outputs without new inference. Text uses the historical Qwen relevance and DeepSeek validity labels. CityLoC contributes three equally weighted seed runs per query. RGB uses a balanced in-map/out-of-map population and the 1 m/2° threshold, including the declared estimated-scale scenes. The maximum possible RGB correct-return rate is 50% and its maximum U_λ is 0.5 in this mixture because out-of-map queries have no correct pose. Rejection receives no positive reward. Consequently, text and RGB panels must not be ranked against each other or pooled into one score. The in-map pose-success rate and out-of-map rejection rate remain separate diagnostics in Table S8.

Observed trade-offs. At $\lambda = 1$, raw QGVI-T yields 16.67 net reward units per 100 queries, compared with -50.00 and -27.96 for the two CityLoC variants. Optional text verification yields 57.22 and dominates the raw route on this first-answer utility for every nonnegative λ: it has both more correct first answers (119 versus 105) and fewer wrong ones (16 versus 75). Its lower Top-5 success and higher latency remain separate trade-offs (Tables S17 and 5). The default raw route's first-answer utility becomes negative when $\lambda > 1.4$; retaining it as the default does not imply superiority when avoiding incorrect answers is the priority.

QGVI-I yields 46.74 net reward units per 100 RGB queries; its 56 wrong returns comprise 43 inaccurate in-map poses and 13 out-of-map accepts. LoGS and SplatHLoc yield -3.48 and -38.36 under their recorded acceptance rules. Equal-scene averaging gives 47.25, -2.75, and -23.83, respectively; all text linear rates and utilities are unchanged because query and seed counts are balanced across scenes. Figure 3 shows pooled and equal-scene utility as the error cost varies. These historical curves retain the native operating points. The completed calibrated comparison adds a distinct-support gate to full LoGS, yielding 46.82 net reward units per 100 queries (53 wrong returns), compared with 46.74 for QGVI-I (56 wrong returns). Its selected threshold is 40; SplatHLoc's retention-constrained count gate selects the native operating point. Table 2 and Supplementary Sec. J give the matched calibration results.

Output budgets and unresolved labels. The shared primary definition uses one answer. Text Top-5 success remains a separate fixed-budget measure: at least one of at most five returned views must be correct. It does not imply that all five views are correct. If L accepted answers are labelled correct and U have unresolved labels, precision lies in $[L/A, (L+U)/A]$ and correct-return rate in $[L/N, (L+U)/N]$. Utility lies in

$$\left[\frac{(1+\lambda)L - \lambda A}{N}, \frac{(1+\lambda)(L+U) - \lambda A}{N}\right]. \quad (8)$$

These are label-uncertainty bounds, not sampling intervals. All accepted answers in Table 1 have resolved historical labels. CityLoC's 540 observations are three runs on 180 query IDs; they are not 540 independent queries. Missing reviewer responses do not become negative labels. Runtime failures count as unsuccessful queries and are reported separately from deliberate abstention; missing execution records invalidate an incomplete evaluation.

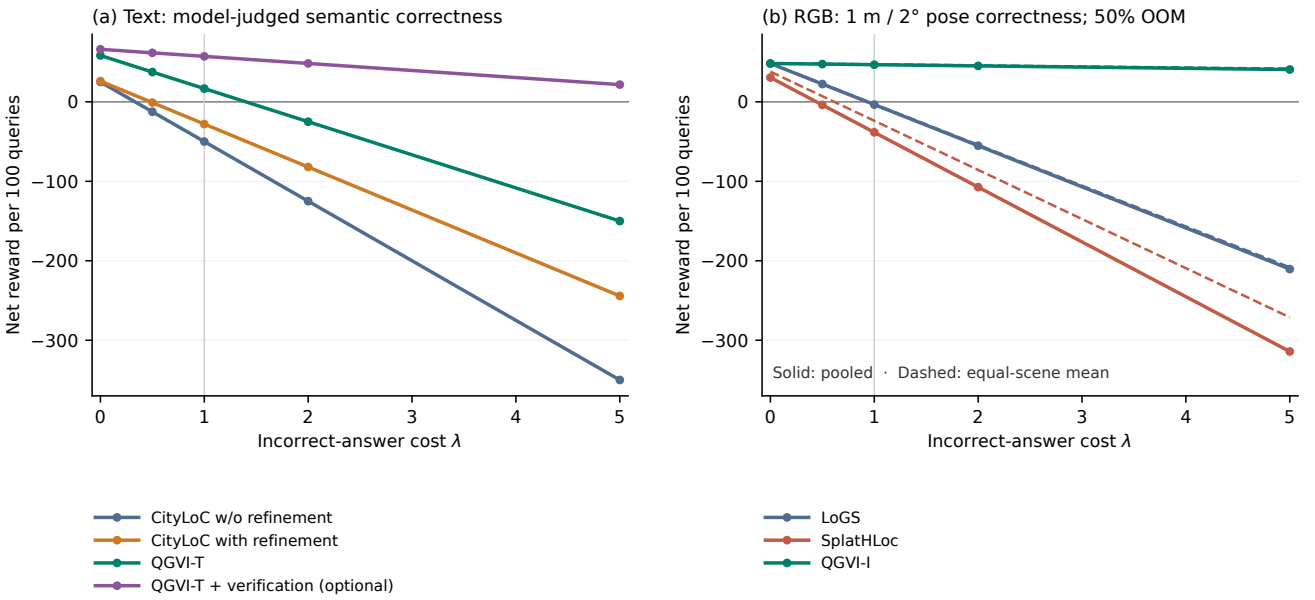


Figure 3. **Sensitivity to the cost of an incorrect first answer.** Correct answers earn $+1$, incorrect answers earn $-\lambda$, and abstentions earn zero. Curves connect the declared cost grid and are linear by definition. Solid curves use pooled query counts; dashed RGB curves weight scenes equally. Text aggregation schemes coincide. The horizontal line is all-reject utility. The two modalities have different correctness definitions and query populations; their heights must not be compared. This historical cost analysis uses native acceptance rules and includes no new inference or threshold optimization; the newly calibrated controls are reported separately in Table 2.

Acceptance and answerability. For a confidence score, thresholds should be selected on validation data under a common retention or risk objective, then frozen for test evaluation. Risk–coverage curves require a recorded score and an explicit threshold policy; isolated binary decisions provide operating points, not a continuous curve. The text cohort does not establish whether each description is answerable anywhere in the scene. We therefore do not credit an empty text return as correct rejection or report text false-accept/false-reject rates. An unsupported source image does not imply that all other views fail. Likewise, Y is not answerability recall, so combining P and Y into a harmonic mean would not produce a conventional F1 score. Full shared risk–coverage curves cannot be recovered from the available operating points: some native abstentions have no returned answer whose correctness can be scored when hypothetically relaxing the policy.

F. Nine-Scene Retrieval and Localization Details

Table S8 retains the per-scene text Top-1/Top-5 success, no-result rates, RGB pose success, false-accept/false-reject rates, and scale factors. These diagnostics complement the pooled first-answer accounting in main Table 1. The two panels use their original query populations and metric denominators; scene means weight all nine scenes equally.

G. Optional Text Verification

QGVI-T + verification evaluates the 20 shortlisted views using a VLM. The verifier receives the rendered image and query, excluding the caption. It enumerates mandatory constraints, including conjunctions, counts, attributes, and spatial relations, and assigns each a label with supporting visual evidence: supported, contradicted, or unclear. A view qualifies only if the verifier reports a match and every reported requirement is supported. Response validation overrides a match verdict when any requirement is contradicted or unclear. Contradiction produces a non-match decision; insufficient evidence produces an uncertain decision.

Eligible views are ranked by verifier relevance, with shortlist order breaking ties. Scores are not calibrated probabilities. We return at most five distinct places with unchanged stored poses, or no result when none qualifies. This abstention concerns the shortlist and does not establish absence throughout the scene. Query processing uses cached renders without new rendering calls.

H. Matched Text Retrieval Controls

CLIP, BM25, MiniLM, and the QGVI-T hybrid use the same 30,522 rendered views, 180 descriptions, and place selection: 64 candidates, a 20-view shortlist, and five distinct returned places. The three caption methods share the same captions; all four omit verification. Each reviewer

Table S8. **Text retrieval and image localization across all nine scenes.** Percentages; scene means weight all nine scenes equally. (a) Joint relevance/validity success on 180 model-judged development descriptions (20 per scene); CityLoC averages three seeds. Empty returns fail. (b) Pose success and false rejection use all in-map queries; false acceptance uses all out-of-map queries. LoGS and SplatHLoc use native maps. QGVI-I gates LoGS initialization without photometric refinement; its gate is calibrated on Block A and SF validation, then fixed. Baselines retain their acceptance rules. † DA3-estimated scale; others source-verified. Scales are fixed across methods and all scenes enter the means. Development results; bold marks the best comparable rate and displayed ties. No-result rates are descriptive.

(a) Text-conditioned view retrieval

Method	Metric (%)	Block Small	Block A	NYC	SF	Citysample	Colosseum	Elvenruin	Park	Road	Scene mean
Queries	n	20	20	20	20	20	20	20	20	20	—
CityLoC [26] without refinement	Top-1 success ↑	21.7	15.0	21.7	43.3	30.0	33.3	20.0	20.0	20.0	25.0
	Top-5 success ↑	25.0	21.7	21.7	45.0	33.3	40.0	23.3	35.0	31.7	30.7
	No result	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
CityLoC [26] with refinement	Top-1 success ↑	21.7	15.0	16.7	48.3	35.0	38.3	20.0	16.7	23.3	26.1
	Top-5 success ↑	25.0	18.3	30.0	56.7	43.3	50.0	25.0	31.7	31.7	34.6
	No result	26.7	35.0	15.0	1.7	41.7	8.3	40.0	8.3	1.7	19.8
QGVI-T	Top-1 success ↑	**65.0**	**60.0**	**60.0**	**55.0**	**50.0**	**50.0**	**80.0**	**40.0**	**65.0**	**58.3**
	Top-5 success ↑	**90.0**	**80.0**	**85.0**	**85.0**	**65.0**	**80.0**	**90.0**	**55.0**	**75.0**	**78.3**
	No result	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

(b) RGB image-query camera localization

Method	Metric (%)	Block Small	Block A	NYC	SF	Citysample	Colosseum	Elvenruin	Park	Road	Scene mean
In-map queries	n	139	602	180	174	142	126	118	228	130	—
Out-of-map queries	n	139	602	180	174	142	126	118	228	130	—
Scale	m / native unit	25	25	1	1	24.777†	12.287†	14.135†	1.040†	1.339†	—
LoGS [8]	Pose 1 m/2° ↑	**90.6**	**93.4**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.3**
	Pose 5 m/5° ↑	**92.1**	**93.5**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.4**
	False accept ↓	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	False reject ↓	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
SplatHLoc [41]	Pose 1 m/2° ↑	65.5	12.6	**100.0**	**100.0**	62.7	82.5	91.5	84.6	84.6	76.0
	Pose 5 m/5° ↑	69.1	18.1	**100.0**	**100.0**	78.9	83.3	94.9	88.6	87.7	80.1
	False accept ↓	100.0	98.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	99.8
	False reject ↓	**0.0**	0.8	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	0.1
QGVI-I	Pose 1 m/2° ↑	**90.6**	**93.4**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.3**
	Pose 5 m/5° ↑	**92.1**	**93.5**	**100.0**	**100.0**	**100.0**	**91.3**	**100.0**	**100.0**	**100.0**	**97.4**
	False accept ↓	**0.000**	**0.332**	**1.111**	**4.023**	**1.408**	**0.000**	**0.000**	**0.000**	**0.000**	**0.764**
	False reject ↓	3.597	2.492	**0.000**	**0.000**	**0.000**	0.794	**0.000**	**0.000**	**0.000**	0.765

judges identical query–image pairs once, using the rubric in Sec. D.2.

I. View-Bank Composition and Novel Views

We vary the candidate observations while retaining the same 180 development descriptions, caption-based scorer, and five-place output budget. The original-image bank contains 18,570 training images, the grid bank contains 30,522 Gaussian renders, and their union contains 49,092 images. All returned cameras are assessed using their Gaussian renderings, including cameras retrieved through original-image captions. Thus the comparison measures semantic access to the supplied asset.

GPT completed all 3,263 distinct query–image judgments for the 64 bank conditions and all 1,329 judgments for the 18 original/union conditions. The following tables use this completed panel. Kimi returned 3,023/3,263 and 1,200/1,329 valid judgments, respectively; the remaining 240 and 129 pairs exhausted the retry budget. These incomplete panels are not used to assert a second-reviewer confirmation of the bank comparisons. Uncertain labels remain unresolved even in the completed GPT panel.

On all nine scenes, adding the grid changes confirmed Top-5 success from 132/180 to 134/180. Its paired change lies in $[-10.0, +12.2]$ percentage points under unresolved labels. This result does not establish a consistent improvement from novel-view augmentation.

Table S9. **Matched text retrieval by scene.** (a) GPT joint relevance-and-validity Top-1/Top-5 success bounds (%), with 20 queries per scene and 180 pooled. GPT completed all 2,825 distinct query–image assessments. (b) Separate pooled Kimi assessment: 2,660/2,825 judgments completed; 165 missing judgments remain unresolved. All methods return five places for every query. Intervals include uncertain labels and, for Kimi, missing judgments; they are not confidence intervals. Bold marks the highest conservative lower endpoint per metric and row in (a), and per metric in (b), including ties. Overlap limits ranking claims.

(a) GPT assessment

Scene	CLIP		BM25		MiniLM		QGVI-T (hybrid)	
	Top-1 ↑	Top-5 ↑	Top-1 ↑	Top-5 ↑	Top-1 ↑	Top-5 ↑	Top-1 ↑	Top-5 ↑
Block Small	35.0–40.0	55.0–70.0	60.0–75.0	70.0–95.0	60.0–70.0	**80.0**–90.0	**65.0**–70.0	**80.0**–90.0
Block A	30.0–60.0	45.0–70.0	**60.0**–85.0	**70.0**–95.0	55.0–65.0	60.0–80.0	50.0–60.0	65.0–85.0
NYC	40.0–55.0	75.0–100.0	**65.0**–80.0	**80.0**–95.0	55.0–75.0	75.0–95.0	**65.0**–85.0	70.0–95.0
SF	60.0–80.0	85.0–100.0	65.0–80.0	90.0–95.0	**80.0**–90.0	90.0–95.0	70.0–90.0	**95.0**–100.0
Citysample	45.0–65.0	60.0–80.0	45.0–75.0	**75.0**–85.0	50.0–75.0	50.0–80.0	**55.0**–80.0	65.0–90.0
Colosseum	40.0–50.0	55.0–85.0	**75.0**–85.0	**85.0**–90.0	55.0–75.0	75.0–95.0	60.0–80.0	80.0–90.0
Elvenruin	50.0–70.0	85.0–90.0	**85.0**–90.0	**95.0**–100.0	75.0–85.0	**95.0**–100.0	**85.0**–95.0	**95.0**–100.0
Park	30.0–40.0	45.0–55.0	**45.0**–60.0	**50.0**–75.0	40.0–65.0	**50.0**–80.0	35.0–55.0	40.0–75.0
Road	25.0–30.0	55.0–75.0	**65.0**–80.0	**80.0**–100.0	40.0–65.0	60.0–95.0	**65.0**–85.0	75.0–100.0
Pooled	39.4–54.4	62.2–80.6	**62.8**–78.9	**77.2**–92.2	56.7–73.9	70.6–90.0	61.1–77.8	73.9–91.7

(b) Kimi assessment, pooled over the same 180 queries

Method	Top-1 (%) ↑	Top-5 (%) ↑	Net utility $100U_1$ ↑
CLIP	32.2–57.2	57.8–90.6	[−35.6, 14.4]
BM25	**55.6**–75.6	**75.6**–96.1	[**11.1**, 51.1]
MiniLM	50.6–76.7	66.1–93.9	[1.1, 53.3]
QGVI-T (hybrid)	52.2–73.9	72.8–93.3	[4.4, 47.8]

Table S10. **Candidate-bank comparison by scene (GPT).** Each scene has 20 queries; pooled rows use all 180. Entries are lower–upper success bounds in percent, not confidence intervals. Top-1 requires a successful first result; Top-5 requires at least one successful result among five. Bold marks the highest confirmed-success lower bound in each scene and metric, including ties; overlapping bounds limit ranking claims. The original bank uses training images, and the union retains those images while adding the grid.

Scene	Original images		Grid renders		Original + grid	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
Block Small	50.0–50.0	75.0–85.0	**65.0**–70.0	**80.0**–90.0	55.0–55.0	75.0–80.0
Block A	40.0–65.0	**70.0**–85.0	**50.0**–60.0	65.0–85.0	45.0–70.0	**70.0**–90.0
NYC	**65.0**–95.0	**75.0**–95.0	**65.0**–85.0	70.0–95.0	**65.0**–90.0	**75.0**–95.0
SF	75.0–85.0	85.0–100.0	70.0–90.0	**95.0**–100.0	**80.0**–90.0	85.0–100.0
City Sample	**55.0**–80.0	**65.0**–90.0	**55.0**–80.0	**65.0**–90.0	**55.0**–80.0	**65.0**–95.0
Colosseum	**65.0**–80.0	**85.0**–90.0	60.0–80.0	80.0–90.0	**65.0**–80.0	80.0–95.0
Elven Ruin	70.0–75.0	80.0–95.0	**85.0**–95.0	**95.0**–100.0	80.0–85.0	**95.0**–95.0
Park	**45.0**–60.0	**55.0**–85.0	35.0–55.0	40.0–75.0	**45.0**–65.0	**55.0**–85.0
Road	55.0–70.0	70.0–90.0	**65.0**–85.0	**75.0**–100.0	50.0–75.0	70.0–90.0
Pooled	57.8–73.3	73.3–90.6	**61.1**–77.8	73.9–91.7	60.0–76.7	**74.4**–91.7

I.1. Appearance and Candidate Coverage

The controlled study uses Block A and SF. A same-pose bank replaces each training image by a Gaussian rendering at the identical camera. The expanded-render bank adds the grid to these training-camera renders. It differs from the nine-scene original-plus-grid bank, which retains the original training images. Table S11 reports both the appearance change and the subsequent coverage change.

I.2. Replacement and Density Controls

Replacement substitutes a fraction of the training-camera renders with grid renders while retaining the original image count and domain counts. Growth retains every training-camera render and adds a fraction of the eligible grid. Both use three fixed geometry-based sampling seeds. NYC and SF have no aerial grid; their original aerial observations remain in the nine-scene comparison. Within the two-scene replacement study, this constraint limits SF to actual re-

Table S11. **Same-pose appearance and expanded-render controls (GPT).** Block A and SF have $N = 20$ queries each; pooled rows have $N = 40$. Same-pose uses Gaussian renders at all training cameras; expanded-render adds grid renders. Bounds are label-uncertainty intervals. Changes are paired Top-5 differences in percentage points against the original-image bank (O) and same-pose bank (R). Bold compares the two rendered-bank conditions within each cohort.

Cohort	Candidate bank	Images	Top-1 (%)	Top-5 (%)	Δ Top-5 vs O	Δ Top-5 vs R
Block A	Same-pose renders	5,138	50.0–65.0	65.0–85.0	[−20.0, +15.0]	[+0.0, +0.0]
Block A	Expanded renders	18,232	**55.0**–75.0	**70.0**–90.0	[−15.0, +20.0]	[−5.0, +15.0]
SF	Same-pose renders	666	**75.0**–85.0	**90.0**–95.0	[−10.0, +10.0]	[+0.0, +0.0]
SF	Expanded renders	3,549	**75.0**–85.0	**90.0**–95.0	[−10.0, +10.0]	[−5.0, +5.0]
Pooled	Same-pose renders	5,804	62.5–75.0	77.5–90.0	[−15.0, +12.5]	[+0.0, +0.0]
Pooled	Expanded renders	21,781	**65.0**–80.0	**80.0**–92.5	[−12.5, +15.0]	[−5.0, +10.0]

placement fractions of 11.9%, 23.7%, and 35.4% at nominal settings of 25%, 50%, and 75%.

All paired Top-5 intervals in these controls include zero. The variation across fractions and seeds does not support a monotonic retrieval benefit from a denser bank. These observations concern the sampled banks and the exposed development descriptions.

I.3. Paired Results by Description Source

The descriptions were authored from 74 held-out original dataset images and 106 novel Gaussian renders. Original dataset images include both real and synthetic source datasets. The source groups have different scene mixtures: Park contributes 18 original-source and two render-source queries, whereas Colosseum contributes one and 19. The source-stratified differences therefore do not isolate a causal effect of query origin.

Confirmed Top-5 successes change from 53 to 52 in the original-source group and from 79 to 82 in the render-source group. Changes in confirmed counts can involve uncertain outcomes, so they need not equal W minus L. Both source groups retain paired uncertainty intervals containing zero.

J. Matched rejection controls and photometric refinement

We apply alternative acceptance rules to the same saved poses and calibrate each rule using only the 490 Block A/SF validation queries (245 in-map and 245 out-of-map). Count thresholds are $\{0, 4, 8, 12, 20, 40, 80\}$; ratio thresholds are $\{0, .05, .1, .2, .35, .5, .7\}$. A feasible rule must retain every pose in each validation-correct set at $1\,\mathrm{m}/2^\circ$ and $5\,\mathrm{m}/5^\circ$. Among feasible rules we maximize out-of-map rejection, then in-map acceptance, and break remaining ties in favor of fewer active gates and a lower threshold. We freeze these rules before scoring the 3,678 queries across nine scenes; the cohort has been used in earlier development and is not a new independent test set. Validation retention does not guarantee retention on these queries.

The raw match ratio divides RANSAC inliers by all saved raw matches; the pair ratio instead uses the deduplicated query-keypoint/landmark pairs supplied to PnP. A unique-keypoint ratio uses distinct inlier query keypoints over distinct matched query keypoints. For refined LoGS, final support is recomputed at the unchanged refined pose using positive depth and a 12-pixel reprojection bound. Initializer-score gates in that panel select already-refined poses and do not save refinement computation. SplatHLoc does not save a correspondence denominator, so we evaluate its observed final inlier count only.

Recalibration selects the current unique-keypoint threshold of 40. Relative to a raw inlier-count threshold of 40 on the same initializer, it removes 11 additional out-of-map returns and two incorrect in-map returns, without losing a correct pose. Applying the distinct-keypoint rule to full LoGS gives 10 false accepts instead of 13 for QGVI-I, at the same 21 in-map rejections and 1,775 strict-threshold successes. Thus the gate also benefits the refined baseline; these results do not establish a rejection advantage over gated full LoGS. For SplatHLoc, a threshold of eight would reject more negatives but lose one validation pose correct at the looser threshold; the no-loss criterion therefore selects its native acceptance point. This is a constraint on the chosen operating point, not evidence that SplatHLoc cannot reject.

Threshold-swept risk–coverage summaries qualify the fixed operating-point comparison. On the identical 3,671 PnP-eligible initializer poses, the grouped right-step area is 0.17638 for unique inlier count, 0.17492 for raw inlier count, and 0.16641 for the unique-keypoint ratio. The unique count therefore has the best reported validation-constrained operating point, but not the lowest integrated risk. These areas cover the observed eligible range only (0–99.81% coverage), without breaking score ties; they are not compared across methods with different poses or eligibility.

Paired photometric diagnostic. We additionally compare the saved initializer with native 20-step LoGS refine-

Table S12. **All replacement and growth settings (GPT).** Every row averages all three fixed seeds on the same 20 queries; seeds do not increase the query denominator. Success columns average the lower and upper label-uncertainty endpoints. The seed range is the minimum–maximum confirmed Top-5 rate. Δ gives mean paired Top-5 uncertainty endpoints against the same-pose bank, in percentage points. For replacement, actual fraction is the proportion of training entries replaced; for growth, it is the proportion of eligible grid entries added. Bold marks the largest mean confirmed rate within each scene and policy.

Scene	Policy	Nominal (%)	Actual (%)	Images	Mean Top-1 bounds (%)	Mean Top-5 bounds (%)	Top-5 seed range (%)	Δ Top-5
Block A	Replace	0	0.0	5,138	50.0–65.0	65.0–85.0	65.0–65.0	[+0.0, +0.0]
Block A	Replace	25	25.0	5,138	55.0–65.0	68.3–85.0	65.0–70.0	[−13.3, +13.3]
Block A	Replace	50	50.0	5,138	**56.7**–70.0	68.3–88.3	65.0–75.0	[−15.0, +20.0]
Block A	Replace	75	75.0	5,138	48.3–61.7	**71.7**–91.7	70.0–75.0	[−13.3, +26.7]
Block A	Grow	25	25.0	8,412	50.0–66.7	68.3–86.7	65.0–70.0	[−5.0, +10.0]
Block A	Grow	50	50.0	11,685	51.7–68.3	68.3–86.7	65.0–70.0	[−6.7, +10.0]
Block A	Grow	100	100.0	18,232	**55.0**–75.0	**70.0**–90.0	70.0–70.0	[−5.0, +15.0]
SF	Replace	0	0.0	666	75.0–85.0	90.0–95.0	90.0–90.0	[+0.0, +0.0]
SF	Replace	25	11.9	666	78.3–85.0	90.0–95.0	90.0–90.0	[+0.0, +0.0]
SF	Replace	50	23.7	666	**83.3**–88.3	90.0–95.0	90.0–90.0	[+0.0, +0.0]
SF	Replace	75	35.4	666	76.7–85.0	**91.7**–98.3	90.0–95.0	[+0.0, +5.0]
SF	Grow	25	25.0	1,387	71.7–83.3	**90.0**–95.0	90.0–90.0	[−3.3, +3.3]
SF	Grow	50	50.0	2,108	71.7–83.3	**90.0**–95.0	90.0–90.0	[−5.0, +5.0]
SF	Grow	100	100.0	3,549	**75.0**–85.0	**90.0**–95.0	90.0–90.0	[−5.0, +5.0]

Table S13. **Paired original-plus-grid versus original-only outcomes (GPT).** W/L/S/U denote confirmed wins, confirmed losses, unchanged outcomes, and unresolved changes. Identical unresolved answer sets have zero paired change and enter S. Bounds are possible paired changes under unresolved labels, not confidence intervals.

Source (N)	Metric	W/L/S/U	Δ (pp)
Original (74)	Top-1	1/1/67/5	[−4.1, +5.4]
Original (74)	Top-5	0/0/64/10	[−9.5, +5.4]
Render (106)	Top-1	4/0/89/13	[−3.8, +12.3]
Render (106)	Top-5	1/1/82/22	[−10.4, +17.0]

ment for every RGB query. All 3,678 pose matrices change. Among the 1,839 in-map queries, strict success remains 1,775 and loose success remains 1,778, with zero individual gains or losses at either threshold. Continuous pose errors nevertheless change: median translation error increases from 0.01191 to 0.02757 m, and median rotation error from 0.02329° to 0.08531°. Translation error improves for 405 queries and worsens for 1,434; rotation error improves for 129 and worsens for 1,710 (a 10^{-9} tolerance in metres or degrees). These aggregate observations on the existing development cohort support omitting this refinement from our evaluated route; they do not show that photometric refinement is generally ineffective. Coordinate preservation applies to the acceptance gate, not to the photometric optimizer.

K. Fixed-Pose Acceptance Audit

Table S16 isolates rejection from pose estimation. Every retained camera matrix is unchanged, and every initializer pose correct at either threshold remains accepted on this cohort. The precision gain therefore comes from removing incorrect outputs, including out-of-map returns, rather than improving pose coordinates.

L. Historical Text Retrieval and Verification Controls

On 176 queries (Table S17(a)), caption retrieval with verification yields 148–155 model-judged Top-5 successes, versus 134–144 for CLIP with the same verifier. Caption retrieval alone yields 142–175; overlapping uncertainty bounds prevent a definitive verifier-effect claim. On 180 queries, verification raises Top-1 success from 58.3% to 66.1% but lowers Top-5 from 78.3% to 72.8%, returning no result for 45/180 queries. This trade-off concerns model-judged development queries.

Table S18 stratifies the 176-query cohort by scene and author-assigned type. Types are not independently grounded in source images; 30 of 34 counting-labelled descriptions lack a target count, so this stratum does not evaluate count constraints.

Table S14. Calibrated RGB rejection controls. All entries pool the nine scenes (1,839 in-map and 1,839 out-of-map queries). FAR and FRR use the out-of-map and in-map denominators, respectively; pose success includes rejected in-map queries as failures. Precision is correct $1\,\mathrm{m}/2^\circ$ poses divided by all accepted queries, including out-of-map returns. $100U_1 = 100(S - W)/3678$. Best values are bold within each LoGS pose family, with ties included; boldface does not imply joint optimality across columns. The SplatHLoc row has its own native poses. † The added threshold is zero because native eligibility already requires four final inliers; thresholds zero and four are equivalent. All gates preserve retained pose coordinates.

Acceptance rule	Threshold	FAR↓ (%)	FRR↓ (%)	Pose success (%)↑ $1\,\mathrm{m}/2^\circ$	Pose success (%)↑ $5\,\mathrm{m}/5^\circ$	Precision↑ (%)	$100U_1$ ↑
Fixed LoGS initializer poses (no photometric refinement)							
Return every pose	—	100.000	**0.000**	**96.52**	**96.68**	48.26	-3.48
PnP success only	—	99.674	0.054	**96.52**	**96.68**	48.35	-3.29
Raw inlier count	40	1.305	1.033	**96.52**	**96.68**	96.26	46.38
Raw match inlier ratio	0.1	79.445	0.761	96.36	96.47	53.93	7.01
Pair inlier ratio	0.1	85.101	0.435	**96.52**	**96.68**	52.27	4.19
Unique-keypoint inlier ratio	0.2	24.035	0.979	96.41	96.57	78.35	34.88
Unique inlier count (refit)	40	**0.707**	1.142	**96.52**	**96.68**	**96.94**	**46.74**
QGVI-I (fixed gate)	40	**0.707**	1.142	**96.52**	**96.68**	**96.94**	**46.74**
Native LoGS refined poses (20 photometric steps)							
Initial raw inlier count	40	1.305	1.033	**96.52**	**96.68**	96.26	46.38
Initial raw match inlier ratio	0.1	79.445	0.761	96.36	96.47	53.93	7.01
Final pair inlier count	40	1.251	1.142	**96.52**	**96.68**	96.41	46.47
Final pair inlier ratio	0.1	81.185	**0.435**	**96.52**	**96.68**	53.40	6.14
Final unique inlier count	40	**0.544**	1.142	**96.52**	**96.68**	**97.10**	**46.82**
Final unique-keypoint ratio	0.2	22.023	0.979	96.41	96.57	79.65	35.89
Native SplatHLoc final poses							
Final inlier count	$0^\dagger$	99.347	0.272	61.17	65.47	30.73	-38.36

Table S15. Scene-level comparison of the calibrated distinct-keypoint gates. Q denotes QGVI-I and L denotes full LoGS with final distinct-inlier count ≥ 40. Both methods have identical pose-success and FRR entries in each scene. Precision and utility use the strict pose threshold. Bold highlights the better Q/L value within each scene, including ties. Estimated DA3 scales for five scenes qualify the metre-based pose thresholds.

Scene	N_{in}	Pose success (%)↑ $1\,\mathrm{m}/2^\circ$	Pose success (%)↑ $5\,\mathrm{m}/5^\circ$	FRR (%)↓	FAR (%)↓ Q	FAR (%)↓ L	Precision (%)↑ Q	Precision (%)↑ L	$100U_1$ ↑ Q	$100U_1$ ↑ L
Block Small	139	90.65	92.09	3.597	**0.000**	**0.000**	**94.03**	**94.03**	**42.45**	**42.45**
Block A	602	93.36	93.52	2.492	**0.332**	**0.332**	**95.42**	**95.42**	**44.44**	**44.44**
NYC	180	100.00	100.00	0.000	1.111	**0.556**	98.90	**99.45**	49.44	**49.72**
SF	174	100.00	100.00	0.000	4.023	**2.874**	96.13	**97.21**	47.99	**48.56**
City Sample	142	100.00	100.00	0.000	**1.408**	**1.408**	**98.61**	**98.61**	**49.30**	**49.30**
Colosseum	126	91.27	91.27	0.794	**0.000**	**0.000**	**92.00**	**92.00**	**41.67**	**41.67**
Elven Ruin	118	100.00	100.00	0.000	**0.000**	**0.000**	**100.00**	**100.00**	**50.00**	**50.00**
Park	228	100.00	100.00	0.000	**0.000**	**0.000**	**100.00**	**100.00**	**50.00**	**50.00**
Road	130	100.00	100.00	0.000	**0.000**	**0.000**	**100.00**	**100.00**	**50.00**	**50.00**

Table S16. **Selective acceptance at fixed poses.** Fixed LoGS geometric estimates, without photometric refinement, are evaluated on 3,678 development queries; the ungated arm includes seven retrieval fallbacks. Pose success uses all 1,839 in-map queries; precision uses all accepted outputs, including distractors. Pooled percentages; the gate changes acceptance only. Bold marks better rates and ties.

Metric	QGVI-I w/o gate	QGVI-I
Accepted outputs (n)	3,678	1,831
False accepts / out-of-map	1,839/1,839	13/1,839
False accept rate ↓	100.000	**0.707**
False rejects / in-map	0/1,839	21/1,839
False reject rate ↓	**0.000**	1.142
Pose success, 1 m/2° ↑	**96.52**	**96.52**
Pose success, 5 m/5° ↑	**96.68**	**96.68**
Precision, 1 m/2° ↑	48.26	**96.94**
Precision, 5 m/5° ↑	48.34	**97.11**

Table S17. **Text component ablations.** (a) Pooled, relevance-only Top-5 success bounds (%) on 176 DeepSeek-authored development queries. (b) Equal-scene joint Top-1 and Top-5 success on 180 queries, with blind Qwen3-VL-8B-Instruct relevance and DeepSeek validity reviews; source support is reviewed separately. Both panels use model judgments. Bold marks the largest lower/upper endpoints separately in (a), where overlap precludes definitive ranking, and highest success in (b).

(a) Retrieval and verification

Retrieval	Visual verifier	n	Relevant Top-5 success (%) ↑
CLIP	No	176	68.2–97.2
CLIP	Yes	176	76.1–81.8
Captions (QGVI-T)	No	176	80.7–**99.4**
Captions	Yes	176	**84.1**–88.1

(b) Joint relevance and validity on the 180-query cohort

Method	Top-1 (%) ↑	Top-5 (%) ↑
QGVI-T	58.3	**78.3**
QGVI-T + verification	**66.1**	72.8
CityLoC, without refinement	25.0	30.7
CityLoC, with refinement	26.1	34.6

Table S18. **Scene and query-type breakdown.** Relevance-only Top-5 success bounds (%) on 176 exposed DeepSeek-judged queries; n counts unique IDs. Categories are author-assigned; answerability and validity are unverified. Of 34 counting descriptions, 30 lack a target count. Bold marks the largest lower/upper endpoints per row, including ties; overlap precludes definitive ranking.

Scene / query type	n	CLIP + verifier Top-5 (%) ↑	QGVI-T + verification Top-5 (%) ↑
Block Small	20	**85.0**	**85.0**
Block A	20	80.0–**85.0**	**85.0**
NYC	20	**90.0**	**90.0**
SF	20	80.0–95.0	**85.0–100.0**
Citysample	20	65.0–70.0	**80.0**
Colosseum	16	75.0–87.5	**87.5–100.0**
Elvenruin	20	60.0–75.0	**75.0–85.0**
Park	20	90.0	**95.0**
Road	20	60.0	**75.0**
Object	61	88.5	**91.8–95.1**
Attribute	35	80.0–85.7	**91.4**
Spatial relation	35	71.4	**85.7**
Counting	34	64.7–**88.2**	**73.5–88.2**
Visible text	11	**45.5**	**45.5**
All queries	176	76.1–81.8	**84.1–88.1**